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Chlorophyll Estimation via Imaging Pipeline Progress Report

Objective: The goal of this image processing pipeline is to be able to measure the chlorophyll concentration ($\mu\text{g}/\text{cm}^2$) of coral samples from scientifically color corrected underwater images. Specifically, this pipeline utilizes distance/depth information alongside the values of the red, green, and blue color channels contained in RAW RGB-D images. With this information, we can calculate both the backscatter and wideband attenuation coefficients with a color chart in the image. We then scientifically correct/restore the colors of the underwater image and analyze the RGB values in each pixel to estimate chlorophyll concentration. This could theoretically be a viable, cost efficient, and non-invasive method of chlorophyll measurement that would bypass the destructive practice of taking physical samples for laboratory analysis.

Current Status: A functioning MATLAB pipeline has been successfully built to process RAW images, but current accuracy is bottlenecked by inconsistent lighting and outdated backscatter/direct signal attenuation estimation methods. The collection of a new dataset with an associated calibration curve and the implementation of modern backscatter/attenuation estimation methods should resolve the current issues. Incorporating accessory pigment considerations may yield novel insights into RGB-D chlorophyll-a estimation methods.

Timeline Overview:

- January 4th: Oculina Imaging and Physical sampling
 - For further information, visit my GitHub notebook:
https://scuba-bo.github.io/BAbramson_Lab_Notebook-Mass_Lab/BA_Training-Log-Oculina-Imaging-&-Sampling-Dive/
- January 5th: Oculina Tissue Removal, Homogenization & Incubation
- January 6th: Chlorophyll Measurement via Optical Density
 - For further information, visit my GitHub notebook:
https://scuba-bo.github.io/BAbramson_Lab_Notebook-Mass_Lab/BA_Training-Log-Oculina-Chlorophyll-&-Surface-Area-Measurement/
- January 11th - 18th: Colorimetry Course
 - Learned how to work with underwater images scientifically
 - Created 3D model of Oculina (sample 1 with strobe lighting only) and generated scaled depth maps
- February 2nd: Surface Area Measurement (Foil method)
- Time until present: Developing pipeline, reviewing literature, troubleshooting, writing progress report

An Idea with Potential/Direction

At this stage, I believe the “red absorption ratio” method of chlorophyll estimation in this pipeline has potential. In Gideon Winter’s article on image based chlorophyll estimation, the red channel is described as having the highest correlation with chlorophyll, with the green and blue channels also being slightly correlated but to a lesser degree than the red channel (Winters et al., 2009). Standard vegetation indices like the Green-Red vegetation index assume that if an organism has more chlorophyll, they will appear more green, but *Oculina patagonica* is a coral with a yellowish-brown color. While chlorophyll-a absorbs red and blue light, the accessory pigment Peridinin absorbs some blue and green light, resulting in the yellowish-brown color (Figure 1). To hopefully account for this, I opted to implement the red absorption ratio ($RAR = G/R$) as a way to measure chlorophyll. If the coral is pale/bleached, the white skeleton will reflect broad-spectrum light relatively equally, meaning the green and red pixel values will be high and fairly similar. Therefore, the G/R ratio will approach 1. In healthy corals with abundant zooxanthellae, the dense Chl-a strongly absorbs red light, drastically reducing the red signal captured by the camera. While peridinin absorbs some green light (Clementson & Wojtasiewicz, 2019; Hochberg et al., 2025), the reduction in green reflectance is less severe than the drop in red reflectance. Consequently, dividing the relatively higher green signal by the heavily attenuated red signal (G/R) should yield a larger ratio. I believe incorporating the red absorption ratio (or other accessory pigment considerations) into the chlorophyll estimation may be a viable way to optimize the chlorophyll measurement from RGB data, rather than just looking at the isolated red channel.

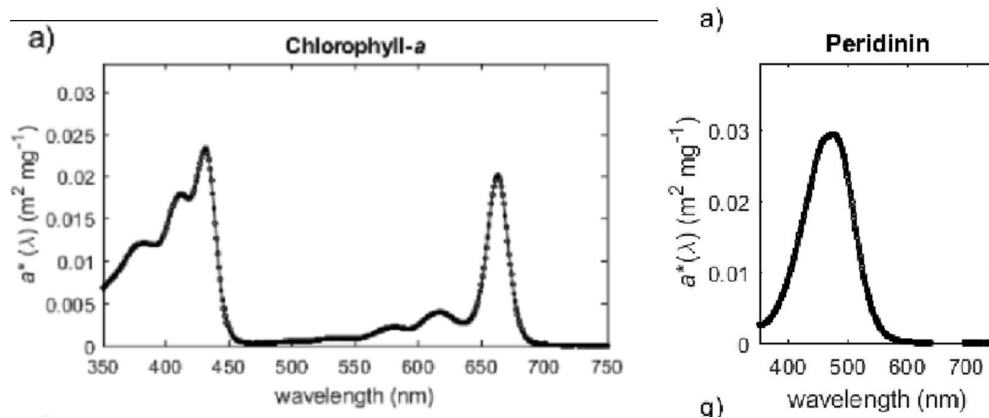


Figure 1: Absorption characteristics of chlorophyll-a and peridinin. Retrieved from “Dataset on the absorption characteristics of extracted phytoplankton pigments” (Clementson & Wojtasiewicz, 2019)

What worked/small victories

While the pipeline development presented several challenges, there were notable technical successes during the initial processing phases. So, I will start this progress report with some small victories from my perspective as an aspiring marine/computer scientist before getting into all the problems.

First of all, one of the first steps to process any of the images is to convert them from raw files into dng files, and then from dng into linear compressed png images (as I learned in the Colorimetry course). The MATLAB script and associated windows executables to accomplish this were provided to me, but my laptop is a mac, so the windows executable was incompatible with my machine. After doing some online research, I managed to compile a mac version of the executables I needed and adjust the MATLAB wrapper scripts to accomplish the same task with my mac. It's not a huge advancement or upgrade, but I learned a lot in the process of troubleshooting this issue.

One aspect of my pipeline that I am quite pleased with is the semi-automatic color chart masking. Instead of manually selecting each of the 18 color patches individually for each image, I implemented a code that allows you to click the four corners of the chart in specific order to collect the same data in a fraction of the time.

While developing the pipeline, I originally was under the impression that I had to calculate the theoretical values of the color chart using the CIE Standard Observer spectral signature csv file. While troubleshooting the errors in my results, I realized that I should instead use the spectral signature of the camera used to take the photos (Canon PowerShot G7 x Mk iii). Fortunately, I was able to conveniently acquire this spectral signature online from a published dataset from the Color Lab ([Spectral Sensitivity Estimation Without a Camera](#)) (Solomatov & Akkaynak, 2023). Swapping the CIE Standard Observer for the unique spectral signature of the camera effectively reduced the angular error between my theoretical values and measurement values.

Primary Obstacles (What didn't work)

Before getting too deep into the results of my work with this data set, I would like to first explain some significant obstacles I encountered. While the overall pipeline structure should function correctly in theory, I believe my backscatter and direct signal attenuation coefficients are intrinsically skewed in this data set for a few reasons.

First of all, the coral images on land were taken with ambient and slightly shaded daytime illumination of the Sdot Yam dive center, but I should have taken the images in direct sunlight. Additionally, I should have taken strobe-lit images on land to compare to the strobe-lit underwater images, but I unfortunately neglected to accomplish this. With my incorrectly illuminated images on land unable to serve as a valid baseline, I opted to compare the restored color charts values to calculated theoretical baseline values based on the known reflectances of

the DGK color chart, the CIE illuminant D65, and CIE standard observer csv files (which I eventually swapped for the spectral signature of the camera).

Another issue I had to contend with was the lack of depth maps for the ambient images. During the Colorimetry course, I created a 3D model and depth maps for the strobe-lit underwater images, but I have not had access to Agisoft to generate depth maps for the ambient underwater coral images. I tried downloading a free trial of the software, but I unfortunately discovered that the feature to generate scaled depth maps is not available in the free trial version. When troubleshooting the lack of ambient depth maps, I realized that I can bypass the need for depth maps by roughly estimating the distance from the camera lens to the chart with the strobe-lit depth maps as a reference. This method allowed me to analyze the ambient underwater images, although any changes made would be applied globally to every pixel regardless of distance from the camera lens. Overall, this workaround allowed me to test the overall pipeline on the ambient images, but it is not an elegant solution. On the topic of 3D models, another issue I encountered was that the color chart was not rigid/secure in the photos. This led to irregular geometry in the 3D models and scaled depth maps.

In the Colorimetry course, I learned how to calculate backscatter by using the grayscale patches of multiple color charts at different distances, but that method didn't smoothly carry over to my current data set with a single color chart. For this data set, there was only one chart and it was quite close to the camera in every photo. When I tried to calculate and subtract backscatter via the grayscale regression method, my results consistently had higher errors. Instead of applying a regression to the grayscale patches to estimate the backscatter, I think it would be worthwhile to instead utilize the augmented dark channel prior method described in the Sea-thru paper which clusters dark pixels together and uses a corresponding depth map to get a more accurate measure of backscatter (Akkaynak & Treibitz, 2019; Kaiming He et al., 2009). The color charts in my images will still be necessary for the creation of scaled 3D models in Agisoft and validation of the color correction algorithm, but as of right now I am using an inadequate method of backscatter estimation. I decided to try a version of the pipeline that ignores backscatter, with my rationale being that since all of the images are very close to the subject, the backscatter should be negligible for this water type. The version of my pipeline ignoring backscatter reduced the angular errors between measured and theoretical values. This helped me pinpoint where some of my issues were, and I realized that my method of backscatter removal was overestimating the backscatter coefficients in my images and stripping them of important data.

Additionally, the lab protocol chlorophyll data I collected (using an H1 Synergy optical density reader) seemed to have some issues due to pipetting (or possibly from other stages in the protocol). Of the two samples (S1 & S4), S1 had a valid chl-a result, but S4 was invalid. Not only that, but the S4 images themselves were strange since the sample was underneath an overhanging ledge which shaded the coral. Overall, the data from S4 was largely unusable, but I was able to experiment with the data from S1.

Finally, the images were not taken with the same exact settings for every image. Later in this report, Table 1 contains all of the images taken and the settings they were taken with. For instance, here my strobe lit underwater images (IMG_2253, Examples 2 & 3) were shot at f/11 and 1/200, but the ambient image in example 1 (IMG_2238) was shot at f/8 and 1/60, while the ambient image in example #4 (IMG_2380) was shot at f/11 and 1/60. Therefore, it's possible to compare examples #2 & #3 to example #4, but example #1 isn't directly comparable due to the shift in aperture settings. From what I understand, changing the aperture setting skews the data, but adjusting the shutter speed should act as a linear scalar that is accounted for with the radiometric calibration in step 5 of the pipeline.

What I will do differently when collecting the next dataset

There are many adjustments I will make when collecting data in the future. First of all, I will maintain the same camera exposure settings across all images (adjusting only shutter speed), and I will bring an underwater slate checklist with me to confirm I collect all the images I need. Additionally, I will take care to secure the color chart more effectively for the sake of accurate geometry in the final 3D model. While imaging, I will focus on sequential overlap between images so that they align properly in Agisoft, and I will also take images from a variety of distances. For example, instead of only taking images close to the coral and the color chart, I will swim about 5 meters away from the chart and then take images as I swim closer. The revised image formation model article discusses this: "choosing coefficients from color charts close to the camera... is a bad habit" and continues "direct signal attenuation coefficients measured close to the camera leads to errors (1m overcompensates, 5m has less errors)" (Akkaynak & Treibitz, 2018). Unfortunately, I only took images quite close to the coral (<1m), so my calculated attenuation coefficients seem to be error-prone.

Furthermore, when I image physical samples on land, I will make sure they are in direct sunlight and not shaded in any way. Additionally, I will take both strobe-lit images alongside ambient images while on land. I also plan to create 3D models (and depth maps) for the images taken under ambient light conditions, rather than only the strobe-lit images, which should provide me with more accurate backscatter and attenuation coefficient calculations. One of the key advancements I hope to implement is the backscatter estimation method described in the Sea-thru paper from 2019 which utilizes dark pixel clustering alongside a depth map instead of the grayscale patch regression method that I used here. Additionally, updating my direct signal attenuation coefficient estimation method to the local space average color (LSAC) method described in the Sea-thru paper should also improve accuracy significantly.

When running physical samples through the laboratory process for chlorophyll measurement, I will ensure I avoid errors while pipetting acetone into the test tubes by practicing with a scale in the lab beforehand. When running the samples through the Synergy H1 optical density reader, I will refrain from discarding any samples until I see the results and check if there were any errors (there is a 1 hour window to run the samples).

Another detail to mention is that in this current pipeline, I set it up in such a way where white is the last patch (#18), but in the csv file with the color chart's reflectances, the white patch is the first patch (#1). In this draft of the pipeline, I kept swapping the csv values to match my numbering system via code, but in the future I will write the pipeline to directly match the csv for practicality.

At this stage, my pipeline is built in a modular fashion for quick troubleshooting adjustments instead of large scale data processing. This unfortunately forces the user to click through all of the steps sequentially in order to get results for a single chart, which is time-consuming. In the future, I will optimize the improved version of the pipeline to be suitable for batch processing large amounts of data. Instead of clicking the corners of a single chart and then going through all of the steps of the pipeline for the individual image, the optimized version will allow the user to click the four corners of a chart within an image and immediately proceed to the next image in the target folder. The pipeline will process the whole batch of images and generate statistics on the results before prompting the user to segment the target coral colonies for chlorophyll estimation. Finally, statistics and plots of the chlorophyll results will be automatically generated for the user alongside a csv file with relevant data.

The calibration curve necessity

In this pipeline, I worked with a single ground truth Chl-a data point ($2.35 \mu\text{g}/\text{cm}^2$) collected from laboratory protocols for chlorophyll extraction. I was able to work with this singular data point by setting the y-intercept to zero, but this forces my chlorophyll estimation section of the pipeline to always shift towards the single data point when attempting to analyze the pixels for chl-a concentration.

Dr. Mass and I recently discussed the next steps of my thesis data collection. We plan to run a new laboratory controlled temperature stress experiment with *Stylophora pistillata* colonies in aquariums. We will steadily increase the temperature over time, and take images/ chl-a during this process. This should allow us to create a new calibration curve with many data points. Hopefully, this preliminary data should be enough for the MSc thesis proposal requirements. Any specific imaging guidance for this upcoming temperature stress experiment would be greatly appreciated.

Pipeline Overview & Breakdown

After taking the raw images, you first need to convert them into dng files, then into compressed png images, and from there you can extract depth data for the complete RGB-D image set (Figure 2). Once you have the RGB-D images, you bring them into MATLAB for the rest of the pipeline steps, which I will explain here. This MATLAB pipeline currently has 9 steps:

- 1) Step 1 of the pipeline is the color chart masking/rgb extraction. You click the four corners of the color charts in a specific order to extract the average rgb values for each patch. Since each square is not the same size, I click in the middle of the longer rectangle patches to ensure the grid aligns with the patches.
- 2) Step 2 calculates the backscatter. Now that we have the locations and rgb values of all patches, we use the depth map to run a linear regression on the greyscale patches against known theoretical reflectances. In the version of the pipeline that ignores backscatter (BA_OculinaPipelineV7_NoBcNoZ.m), this step sets a fixed global distance and sets backscatter to 0.
- 3) Step 3 is the backscatter subtraction step. A saturating exponential physical model is fit to the backscatter data, creating a “backscatter map” and subtracting it from the original photo. In the version of the pipeline ignoring backscatter, this step does not alter the image.
- 4) Step 4 is the wideband attenuation coefficient calculation. The direct signal of the 25% grey patch (#17 in my chart ordering system) is compared to its theoretical target values to determine the rate of light attenuation at that depth.
- 5) Step 5 applies attenuation correction & radiometric calibration to the pixels, restoring the image’s colors based on the attenuation coefficients calculated in step 4. The brightness is radiometrically calibrated to match the theoretical values.
- 6) Step 6 is the calculation of theoretical true XYZ values of the color patches. This incorporates data such as the reflectances of the D50 color chart, the CIE standard illuminant D65, and the CIE standard observer. A color correction matrix is calculated and applied to the image, with the overall fit of the matrix plotted on a graph as well. (I eventually swapped the CIE standard observer data for the unique spectral signature of the camera used to take the images (Canon G7 x mk iii), and the script with the canon signature is named BA_OculinaPipelineV7_NoBcNoZ_Canon.m.)
- 7) Step 7 is the calculation of angular error between the measured XYZ values and theoretical XYZ values. The average is calculated based on all 18 patches.

All of the above steps (1→ 7) are shown conceptually in the flowchart Figure 4. The chlorophyll estimation represented in steps 8 & 9 are shown conceptually in Figure 5.

- 8) Step 8 of the pipeline prompts you to draw a polygon mask to segment coral tissue from the rest of the image, calculates the “red absorption ratio” (green/red) to map chlorophyll-a, and scales it linearly to the known laboratory chlorophyll-a concentration ($2.35 \mu\text{g}/\text{cm}^2$) to plot a heatmap for chlorophyll.
- 9) Finally, step 9 extracts the mean, median, and standard deviation of the valid pixels within the chlorophyll heat map and plots a bar chart with error bars. It also exports the metrics into a csv file.

Table 1: Image settings for Oculina sample 1. All images were taken in manual mode with ISO 200 and White balance (K) of 5400.

Photographer	Location	Lighting	# of photos	File names	aperture	shutter speed	Removed Images	Depth Maps
Hagai	UW	Strobe	20	2217 --> 2237	f/11	1/160		No
Hagai	UW	Ambient	13	2238--> 2251	f/8	1/60		No
Boaz	UW	Strobe	20	2253 --> 2273	f/11	1/200	2257 (glare)	Yes
Boaz	UW	Ambient	15	2280 --> 2295	f/11	1/60	2274 (dark, f/11/ 1/200), 2275 (marker), 2276-->2278 (f/8, 1/80), 2279 (marker)	No
Boaz	Land	Ambient (shaded)	8	2362 --> 2370	f/9	1/60		No

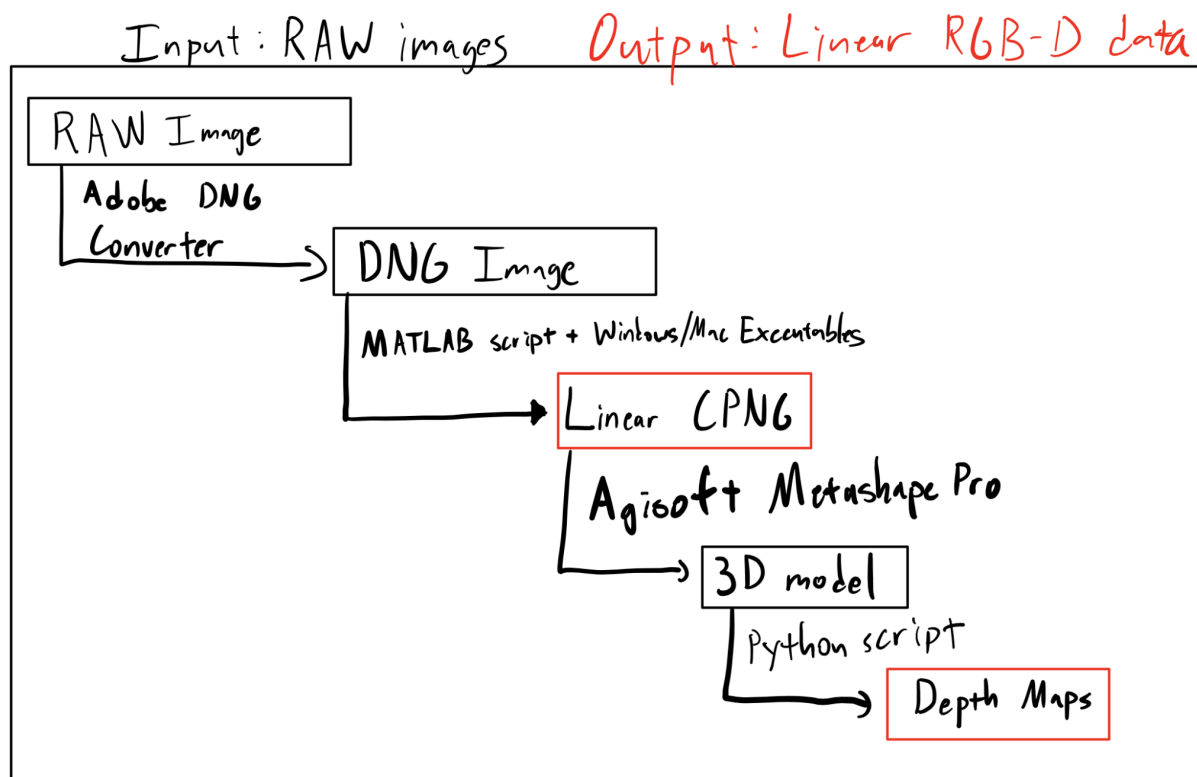


Figure 2: RAW Image to linear RGB-D image conversion process schematic.

Revised Underwater Image Formation Model:

$$\text{Camera image (input)} \rightarrow I_c = \underbrace{J_c e^{-B_c^D(v_D)z}}_{\substack{D_c \\ \text{Direct Signal}}} + \underbrace{B_c^\infty (1 - e^{-B_c^B(v_B)z})}_{\substack{B_c \\ \text{Backscatter}}} \rightarrow \text{Restored image (output)}$$

Figure 3: Revised Underwater Image Formation Model (Akkaynak & Treibitz, 2018). I_c represents the image captured by the camera underwater, J_c represents the unattenuated direct signal (what the image would look like without the water), B_c represents the backscatter introduced to the captured image. Vectors v_D and v_B represent the coefficient dependencies $v_D = \{z, \rho, E, S_C, \beta\}$ and $v_B = \{E, S_C, b, \beta\}$.

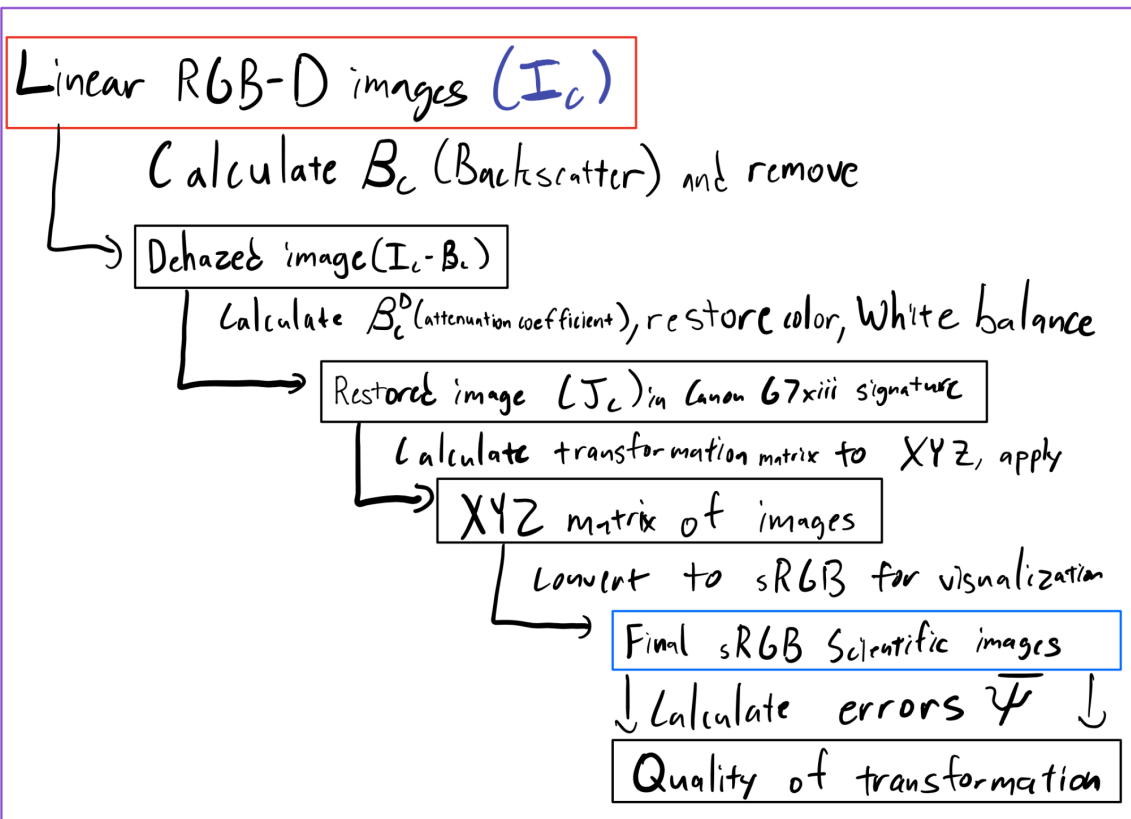


Figure 4: Flow chart depicting the process of converting linear RGB-D images to scientifically reproducible images.

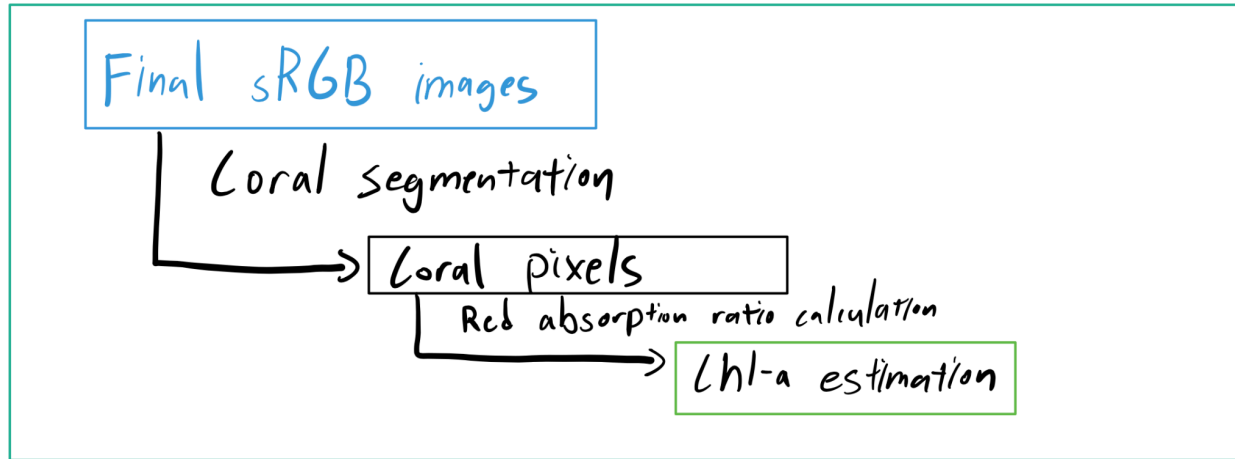
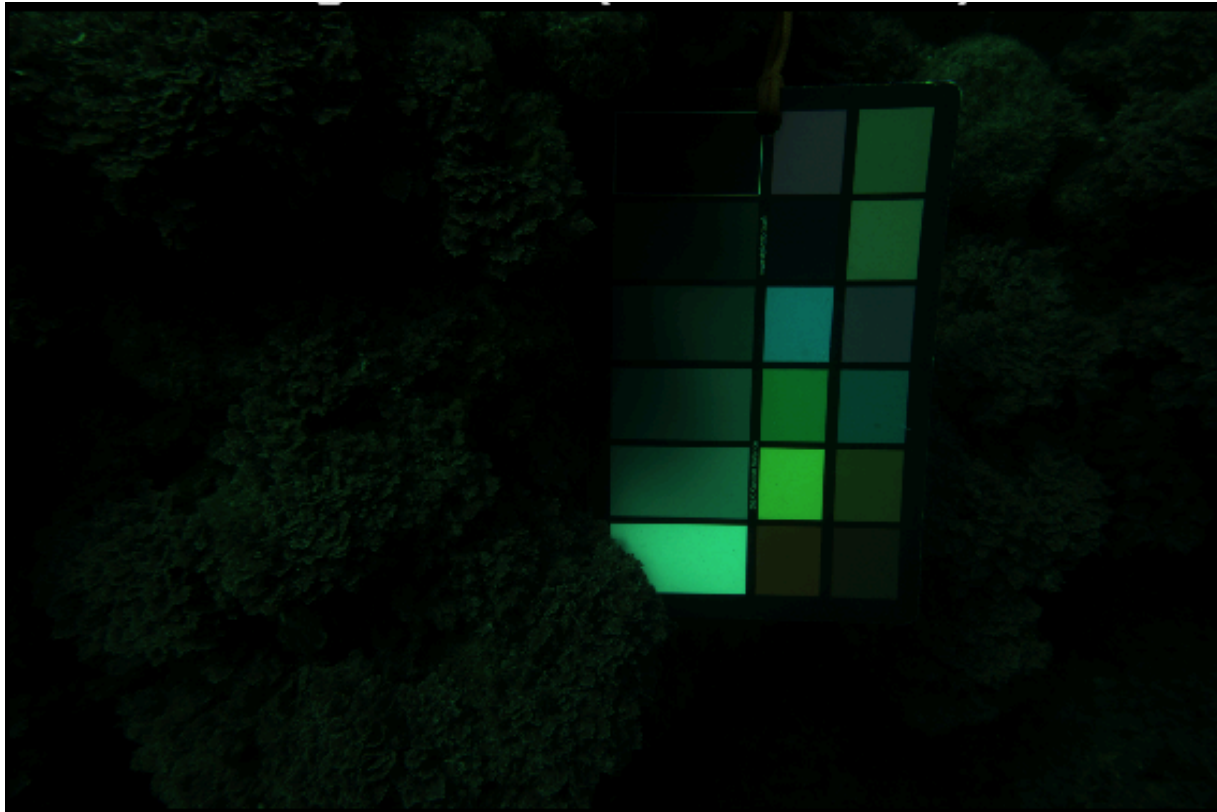


Figure 5: Flow chart of the chlorophyll-a concentration estimation via coral sample segmentation and pixel analysis.

Table 2: Overview of the three example pipeline runs shown in this report. Note the increase in error in example 2, which uses grayscale regression for backscatter estimation.

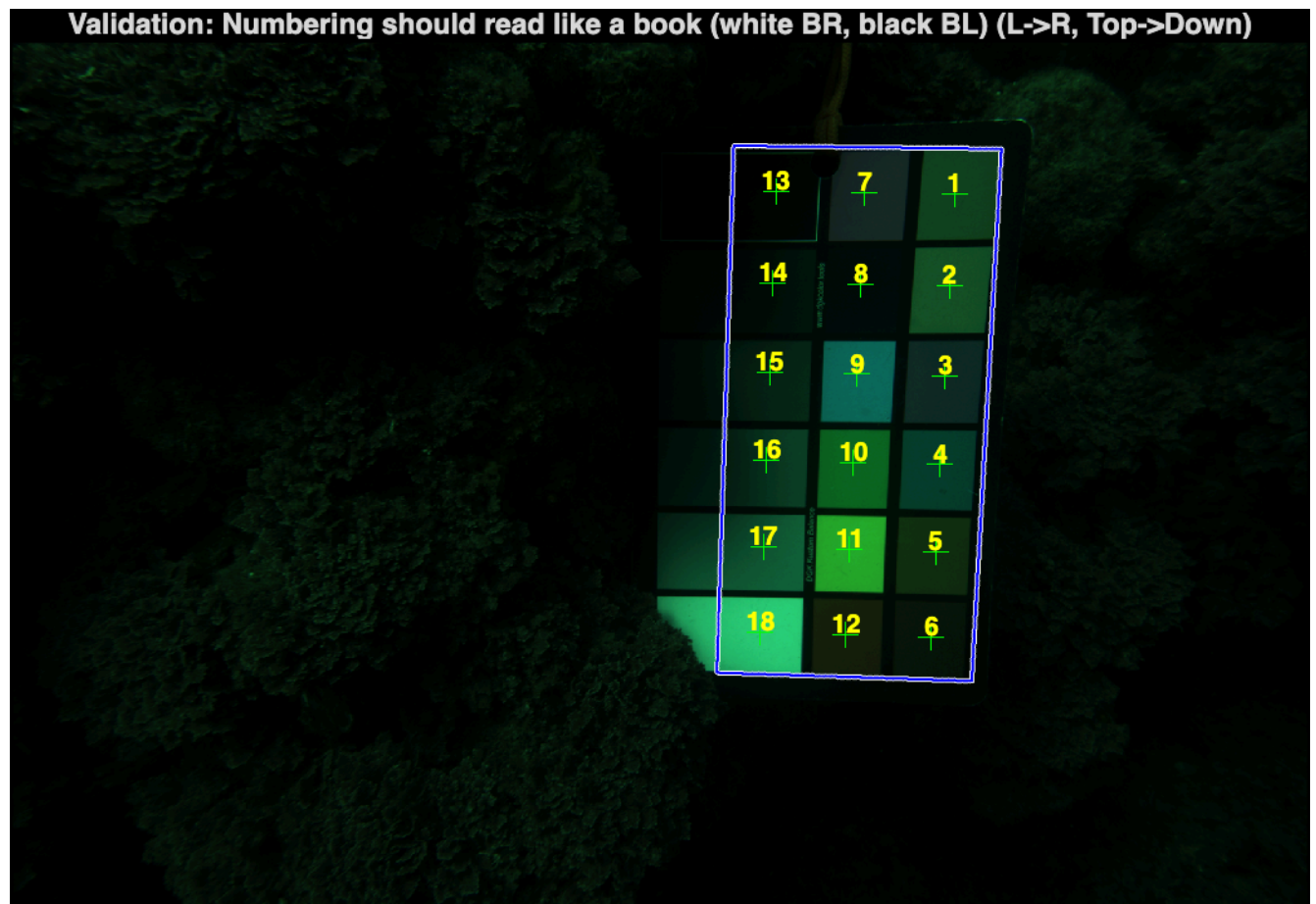
	Example 1:	Example 2	Example 3	Example 4
File Name	IMG_2238	IMG_2253	IMG_2253	IMG_2280
Lighting	Ambient	Strobe-lit	Strobe-lit	Ambient
Camera Settings	f/8 & 1/60	f/11 & 1/200	f/11 & 1/200	f/11 & 1/60
Script Used:	BA_OculinaPipelineV7_NoBcNoZ.m	BA_OculinaPipelineV7_YesBcYesZ.m	BA_OculinaPipelineV7_NoBcNoZ.m	BA_OculinaPipelineV7_NoBcNoZ_Canon.m
Backscatter:	Assumed negligible	Calculated via grayscale regression and subtracted	Assumed negligible	Assumed negligible
Reference Theoretical CSV	CIE Standard Observer	CIE Standard Observer	CIE Standard Observer	Canon G7 x mk iii
Angular Error:	10.9 degrees	18.88 degrees	11.98 degrees	10.55 degrees
Chl-a estimation:	2.35 ug Chl/cm ² , Std. Dev. =0.3499	Mean = 2.35 ug Chl/cm ² , Std. Dev. =2.5217	Mean = 2.35 ug Chl/cm ² Std. Dev. =0.3134	Mean = 2.35 ug Chl/cm ² Std. Dev. = 0.3845

Example #1: IMG_2238



- Lighting: Ambient (No strobe)
- Script Used: BA_OculinaPipelineV7_NoBcNoZ.m
- No depth map, assumed distance from camera to chart and coral was 0.4 meters (based on similar images that had depth maps).
- Assumed backscatter was negligible.
- Used theoretical/calculated land baseline color patch values for the XYZ conversion instead of the images on land since the color chart was shaded in the land images. Theoretical values incorporated the reflectances csv, CIE standard illuminant D65 csv, and standard observer csv files for the calculation.
- Final results: **Average angle error = 10.9 degrees**
- The calibration curve had only a single input from laboratory protocols (2.35 ug Chl/cm²), so the y intercept was set to zero and the line drawn between the two points operates as the calibration curve until more robust data is gathered.
- Chlorophyll estimation: **Mean = 2.35 ug Chl/cm², Std. Deviation = 0.3499**

Ex. 1 - Step 1: Extract Color Chart Data & Create Masks



Ex. 1 - Step 2: Global Depth Value Assignment & Assume Backscatter Negligible

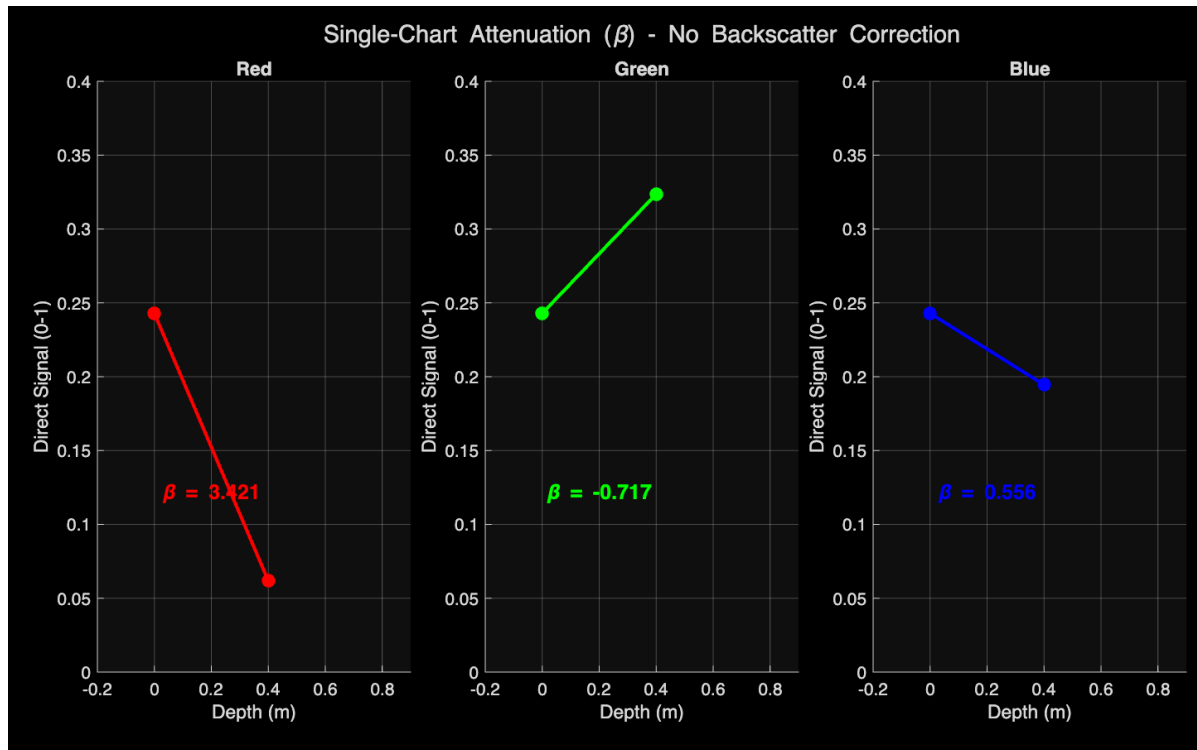
- Normally this step would obtain the depth values from the depth maps and calculate backscatter using the grayscale regression, but in this case I did not have a depth map. Instead, I assumed the chart and coral were 0.4 m from the camera lens and I also chose to assume that backscatter was negligible.

Ex. 1 - Step 3: Backscatter Modeling and Subtraction

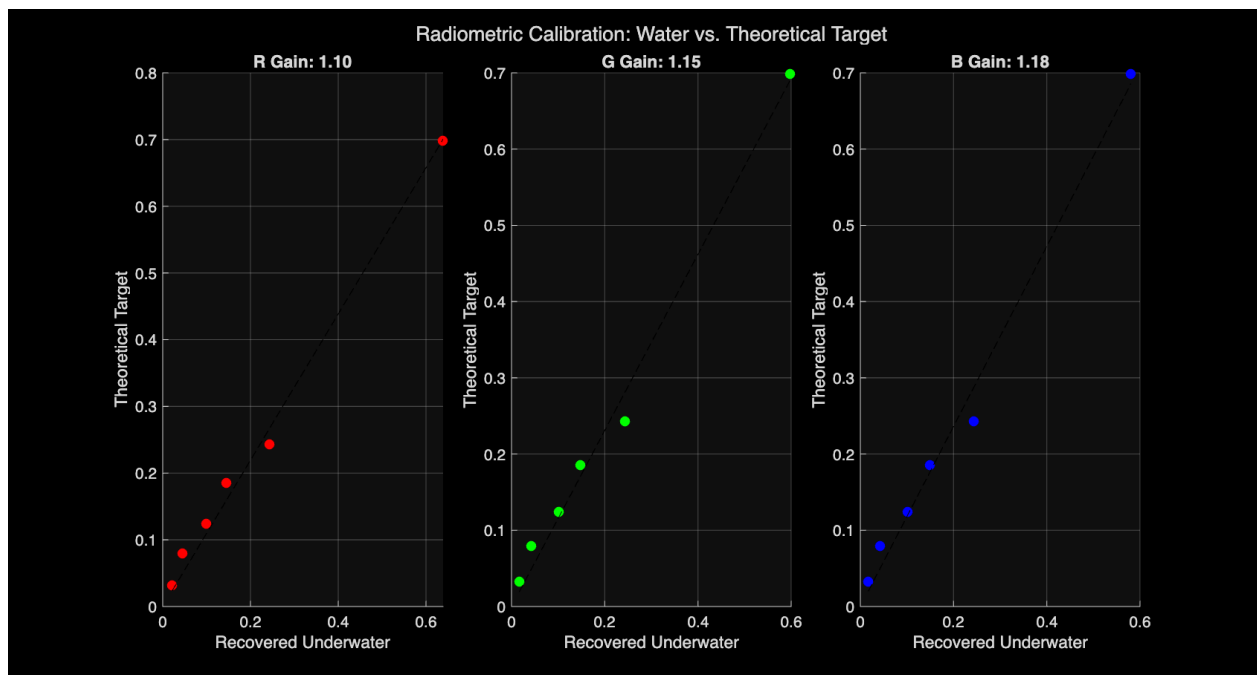
- Once again, in this scenario I assumed backscatter to be negligible so this step in the script doesn't change the image.

Ex. 1 - Step 4: Wideband Attenuation Estimation (Beta) - Theoretical Method

- Calculates color attenuation (Beta) assuming negligible backscatter.
- Uses Eq 9 from Sea-thru paper with W_c extracted from the 25% gray patch (Patch 17).
- Instead of a land image, it uses the theoretically calculated values.
- $\text{Beta} = -\ln(Dc_patch17 / \text{Theoretical_patch17}) / z$



Ex. 1 - Step 5: Attenuation Correction & Radiometric Calibration (Theoretical)

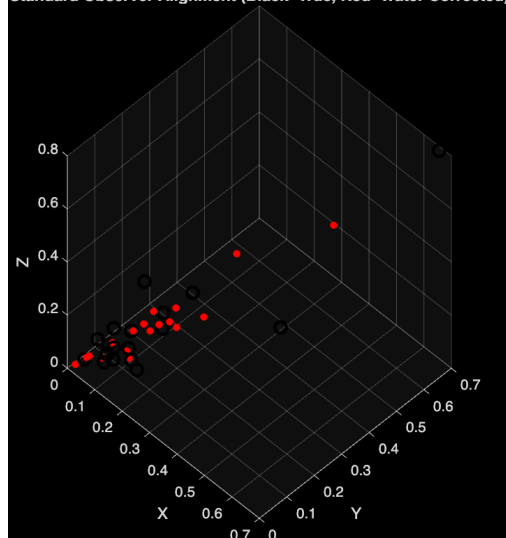




Ex. 1 - Step 6: Universal XYZ Transformation (Water-Corrected -> Standard Observer)

- Calculates 3x3 matrix mapping the water-corrected RGB directly to CIE XYZ (D65).
- This version calculates the target XYZ mathematically from first principles using the actual spectral reflectances, standard observer, and illuminant D65.

Standard Observer Alignment (Black=True, Red=Water Corrected)

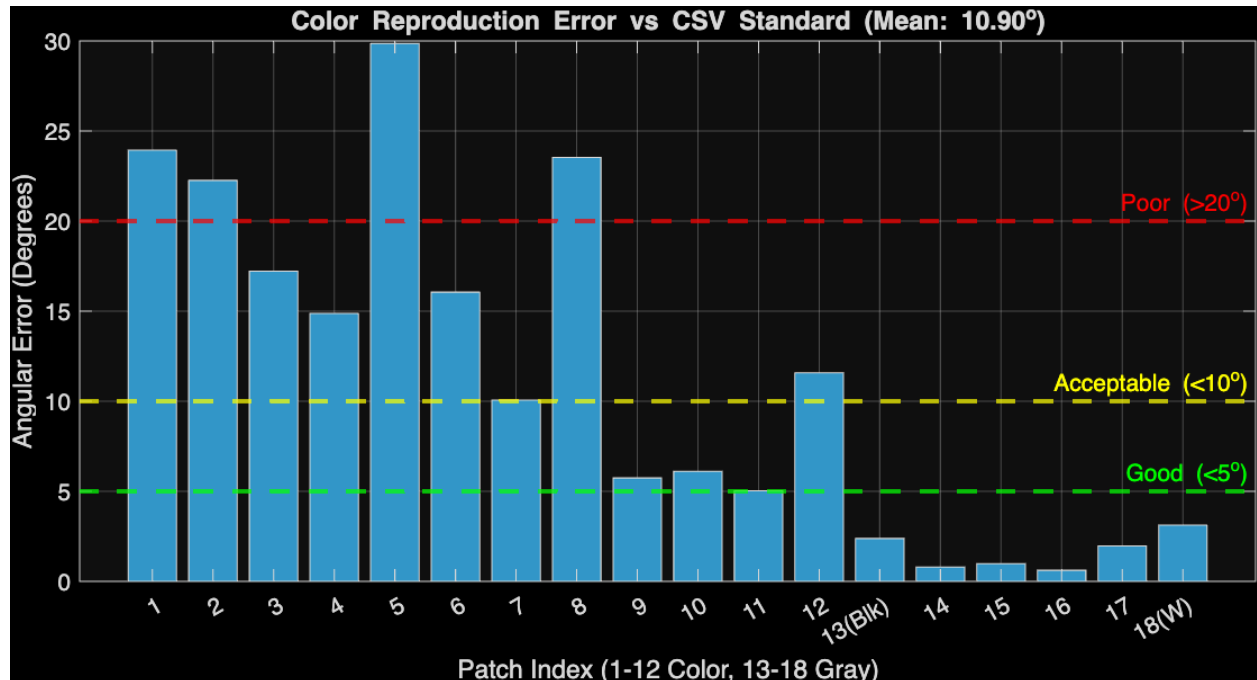


Final Scientific Image (sRGB Preview)



Ex. 1 - Step 7: Color Consistency Error Analysis (Theoretical XYZ)

- Calculates the angular error (in degrees) between the final corrected underwater XYZ and the true CSV XYZ standard.



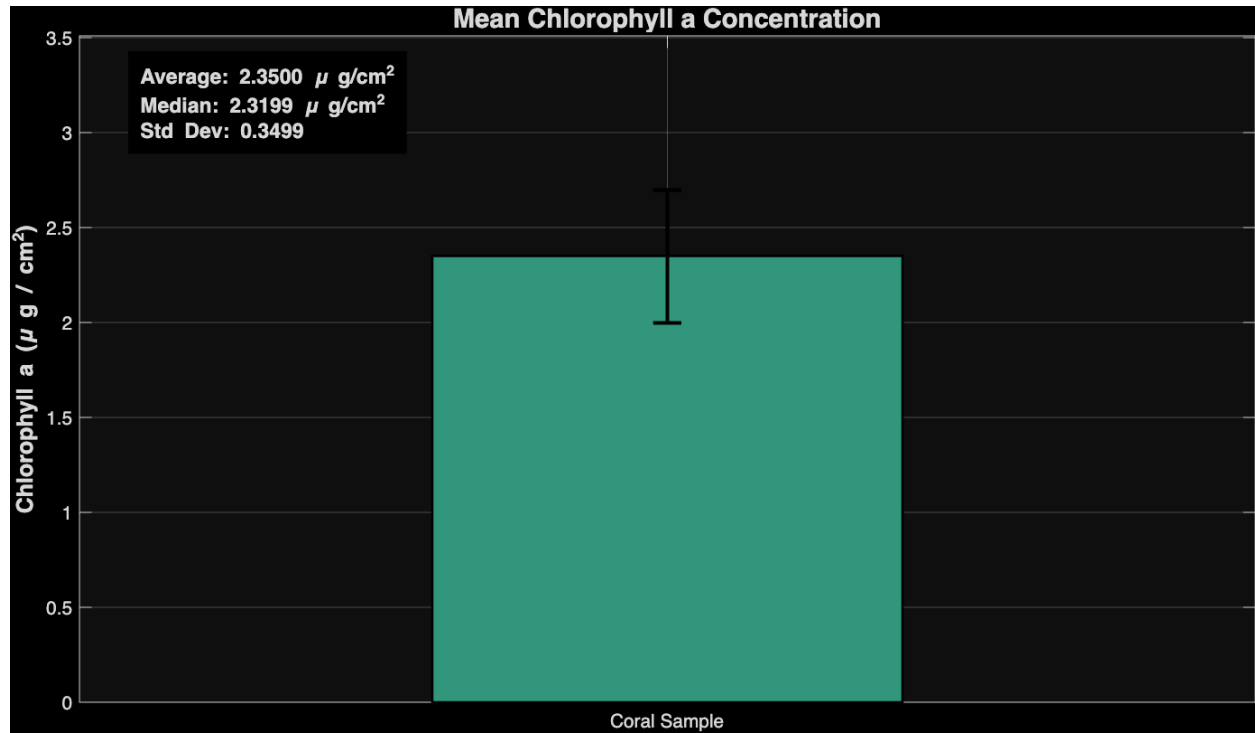
Ex. 1 - Step 8: Coral Masking and Single-Point Chlorophyll Estimation

- Isolates the coral from the background and applies a single-point proportional scaling to map chlorophyll concentration.



Ex. 1 - Step 9: Chlorophyll Data Extraction & Plotting

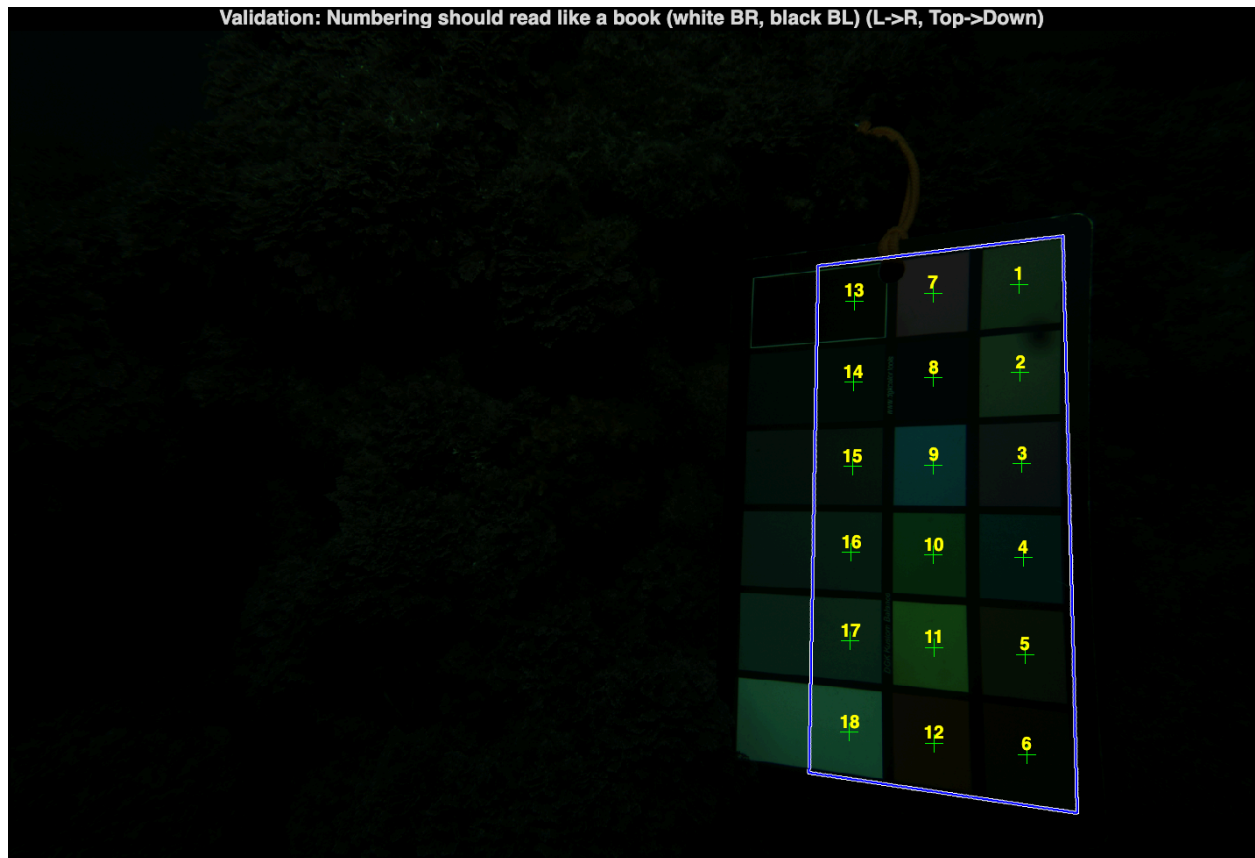
- Extracts the average chlorophyll values from the generated map and visualizes the data with error bars and text.



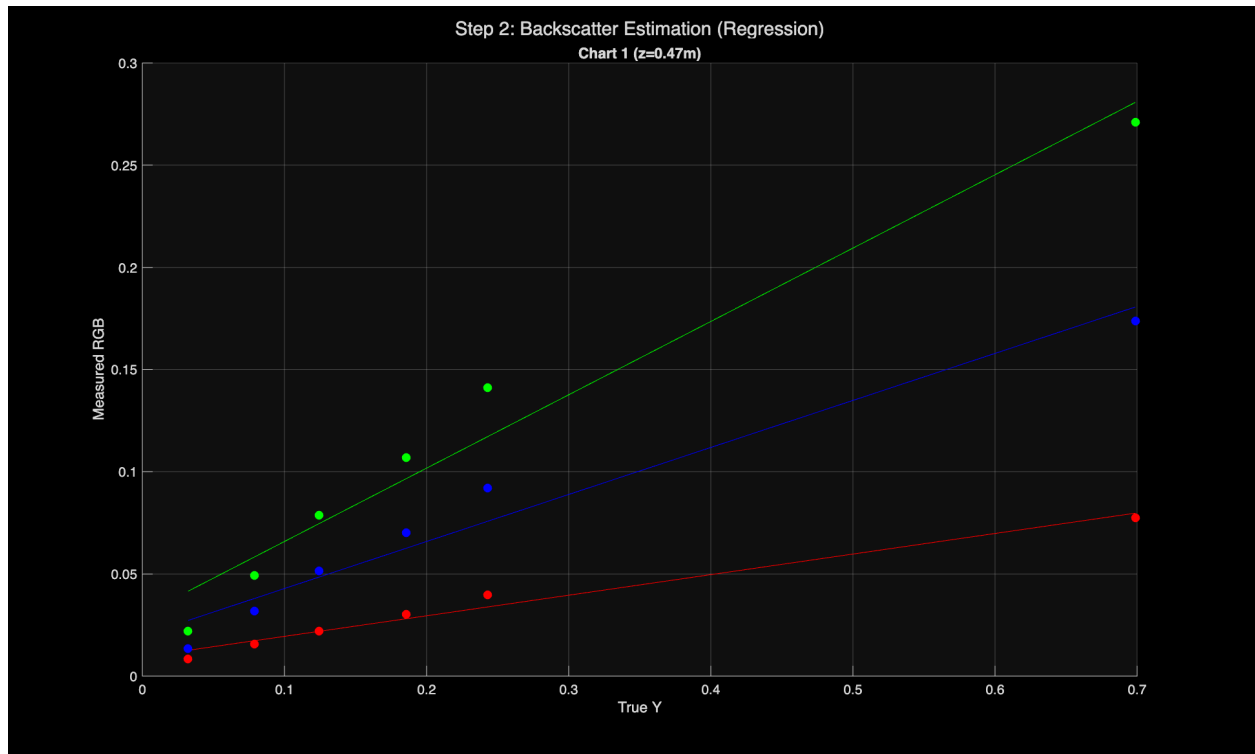
Example 2: IMG_2253 (Backscatter removed)

- Lighting: Strobe-lit
- Script: BA_OculinaPipelineV7_YesBcYesZ.m
- Depth map available & utilized.
- Backscatter was calculated and removed via grayscale regression.
 - Example #3 uses the same image but assumes backscatter to be negligible.
- Used calculated theoretical XYZ values for the transformation matrix. Theoretical values incorporated the reflectances csv, CIE standard illuminant D65 csv, and standard observer csv files for the calculation.
- Final results: **Average angle error = 18.88 degrees**
- Chlorophyll estimation: **Mean = 2.35 ug Chl/cm², Std. Deviation = 2.5217**

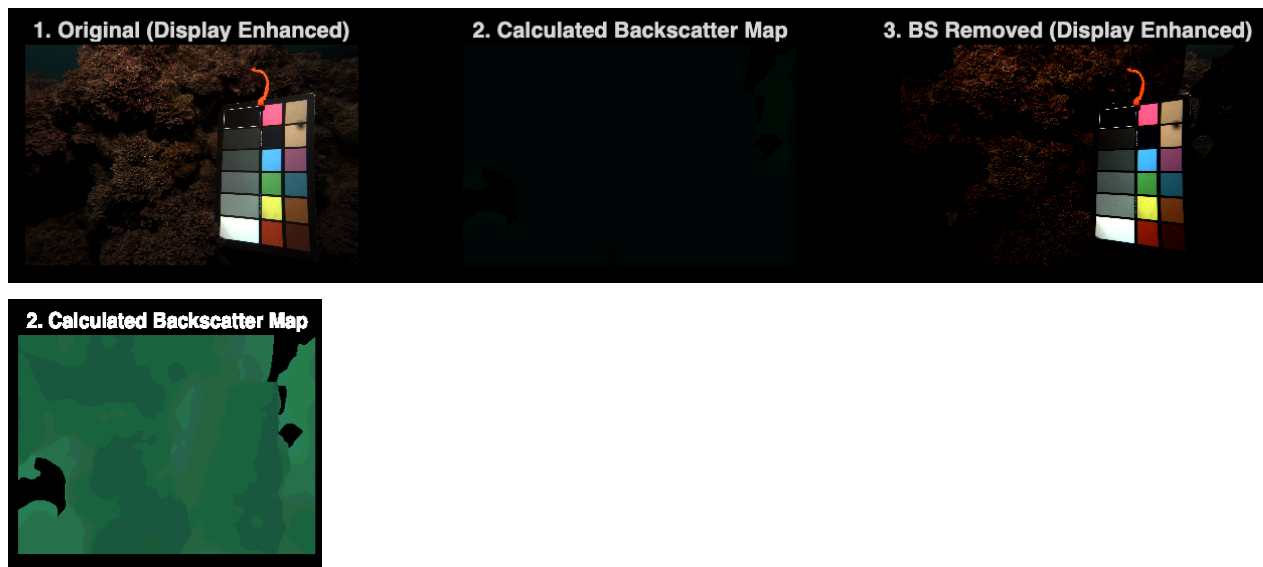
Ex. 2 - Step 1: Extract Color Chart Data & Create Masks



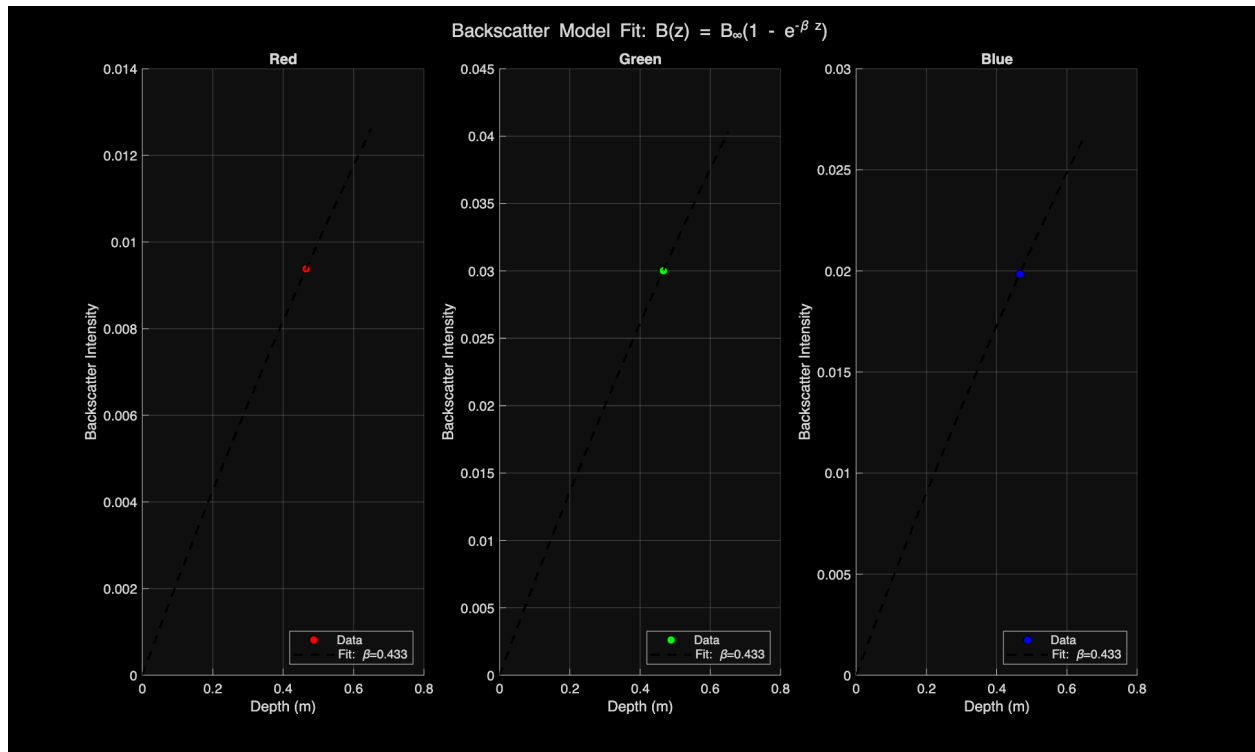
Ex. 2 - Step 2: Depth Extraction & Backscatter Estimation (grayscale regression method)



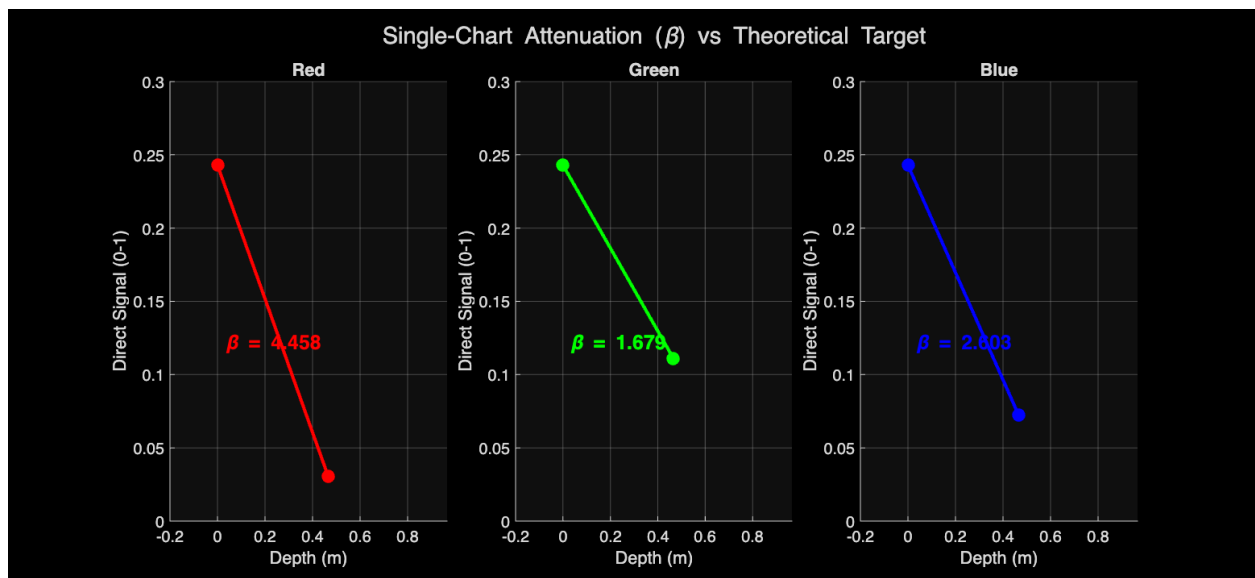
Ex. 2 - Step 3: Backscatter Modeling & Subtraction



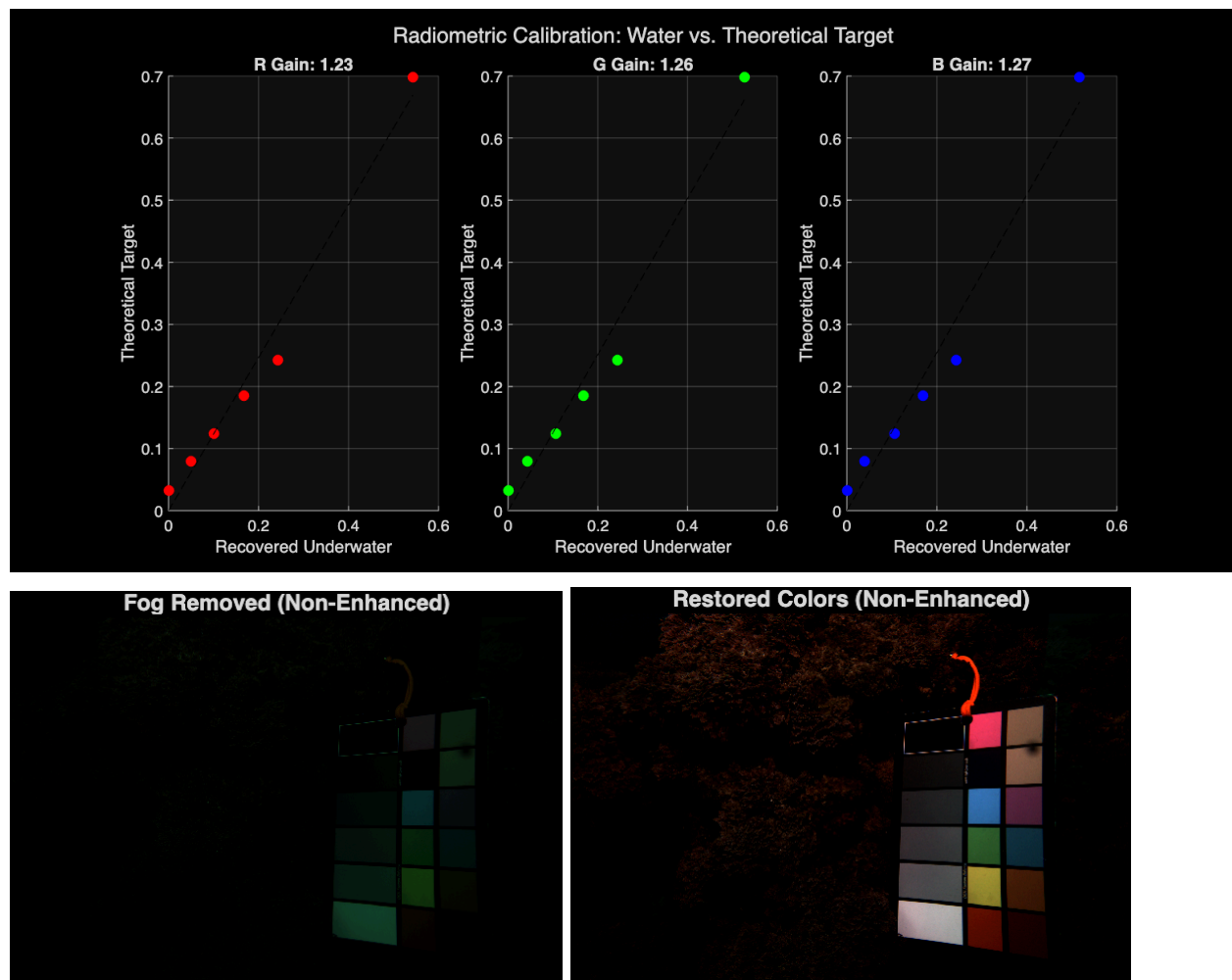
➤ Brightened backscatter map for visualization



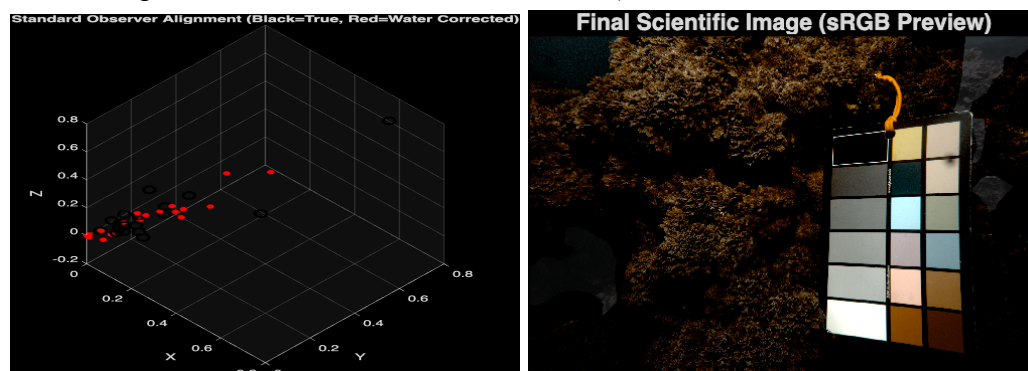
Ex. 2 - Step 4: Wideband Attenuation Estimation (Beta) - Theoretical Method



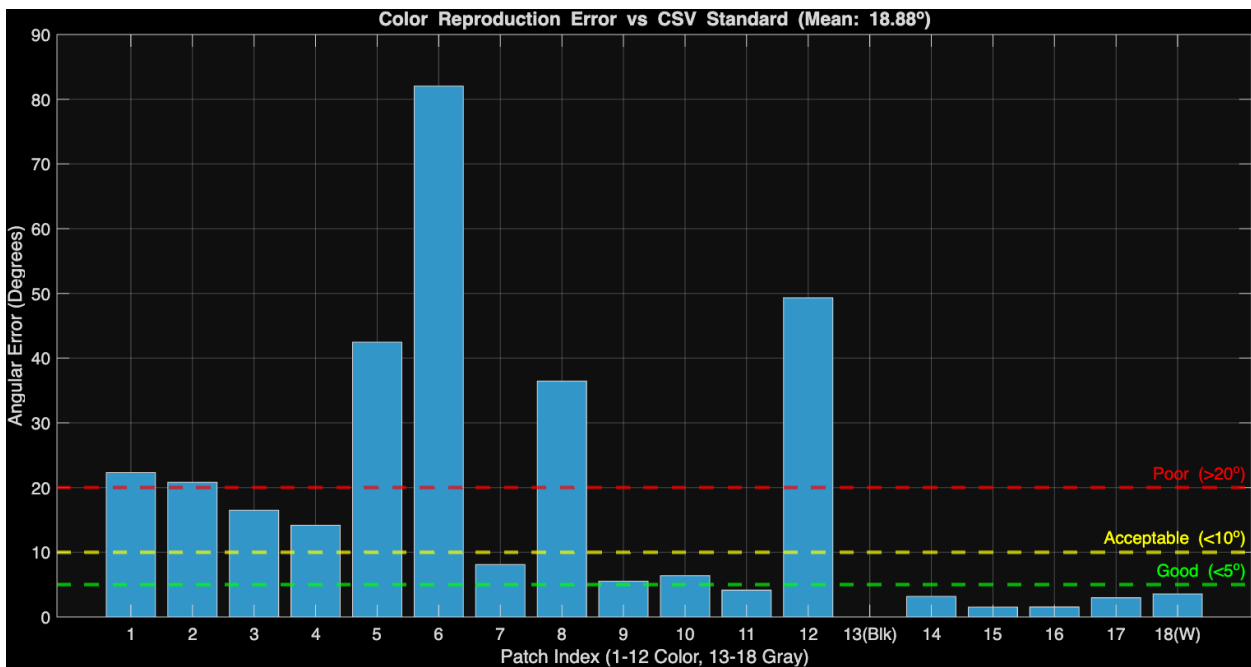
Ex. 2 - Step 5: Attenuation Correction & Radiometric Calibration (Theoretical)



Ex. 2 - Step 6: Universal XYZ Transformation (Water-Corrected -> Standard Observer)



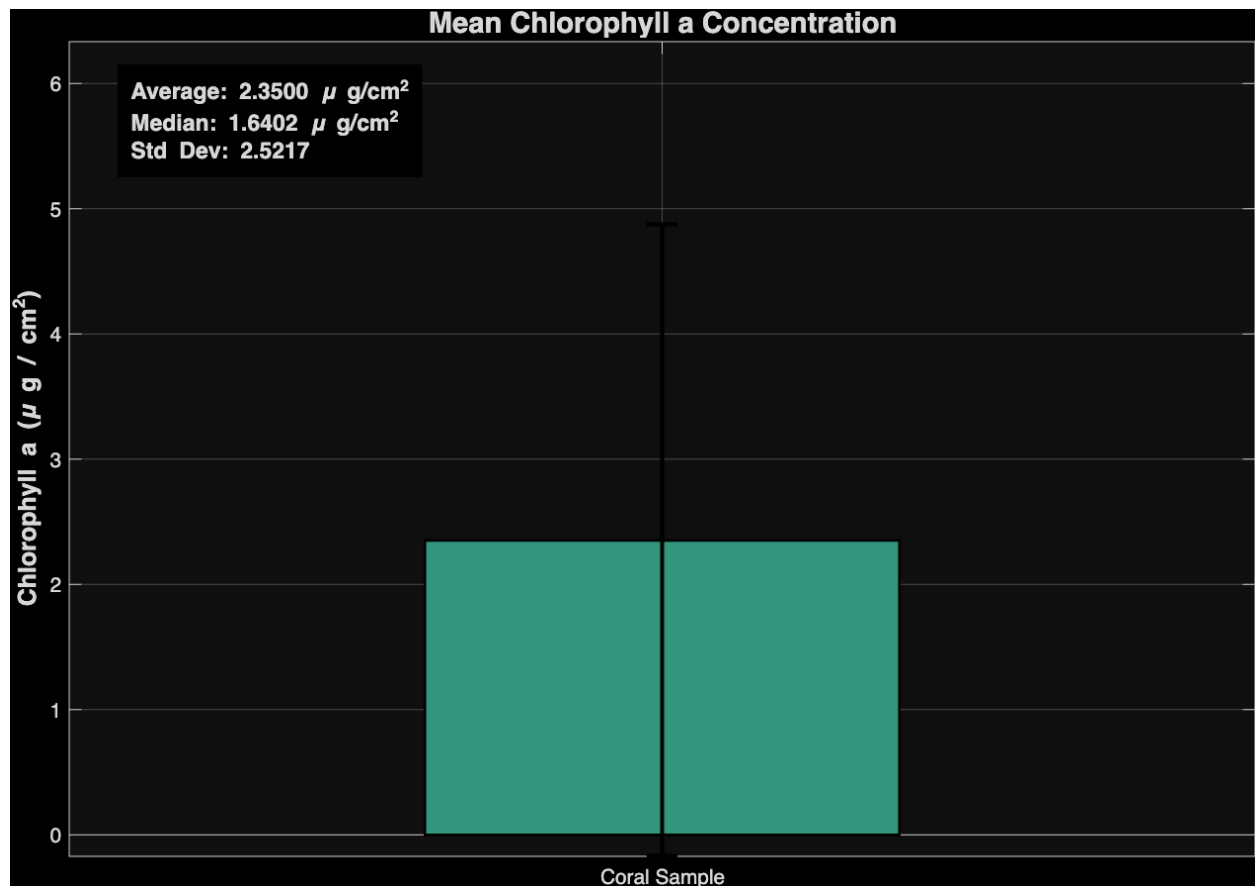
Ex. 2 - Step 7: Color Consistency Error Analysis (Theoretical XYZ)



Ex. 2 - Step 8: Coral Masking and Single-Point Chlorophyll Estimation



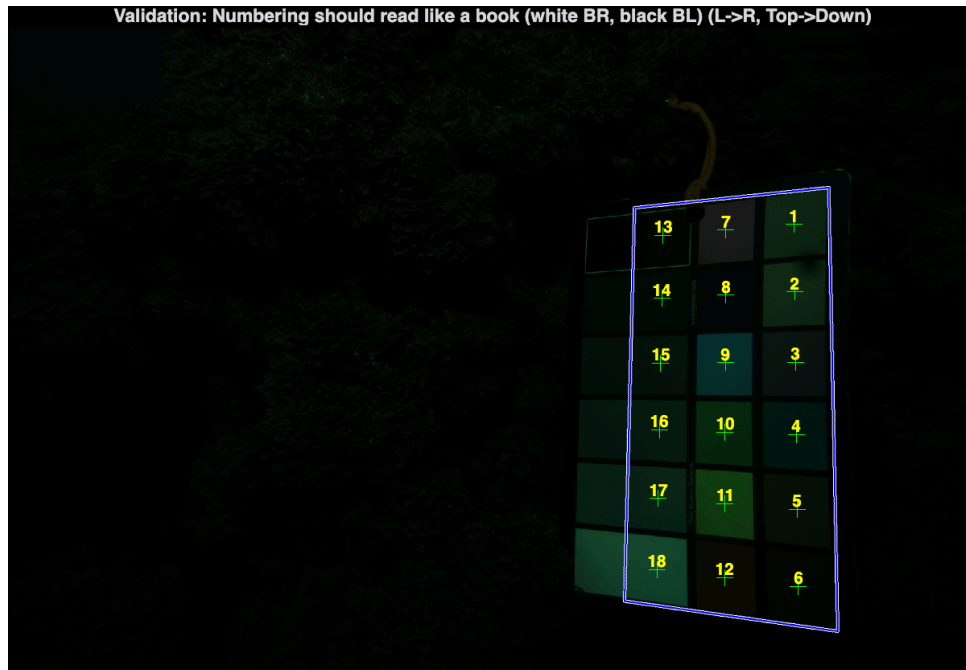
Ex. 2 - Step 9: Chlorophyll Data Extraction & Export



Example #3: IMG_2253- same as Example #2, but backscatter is assumed to be negligible

- Lighting: Strobe-lit
- Script: BA_OculinaPipelineV7_NoBcNoZ.m
- Depth set globally to 0.47 m
- Backscatter assumed to be negligible.
- Used calculated theoretical XYZ values for the transformation matrix. Theoretical values incorporated the DGK reflectances csv, CIE standard illuminant D65 csv, and standard observer csv files for the calculation.
- Final results: **Average angle error = 11.98 degrees**
- Chlorophyll estimation: **Mean = 2.35 $\mu\text{g Chl}/\text{cm}^2$, Std. Deviation = 0.3134**

Ex. 3 - Step 1: Extract Color Chart Data & Create Masks



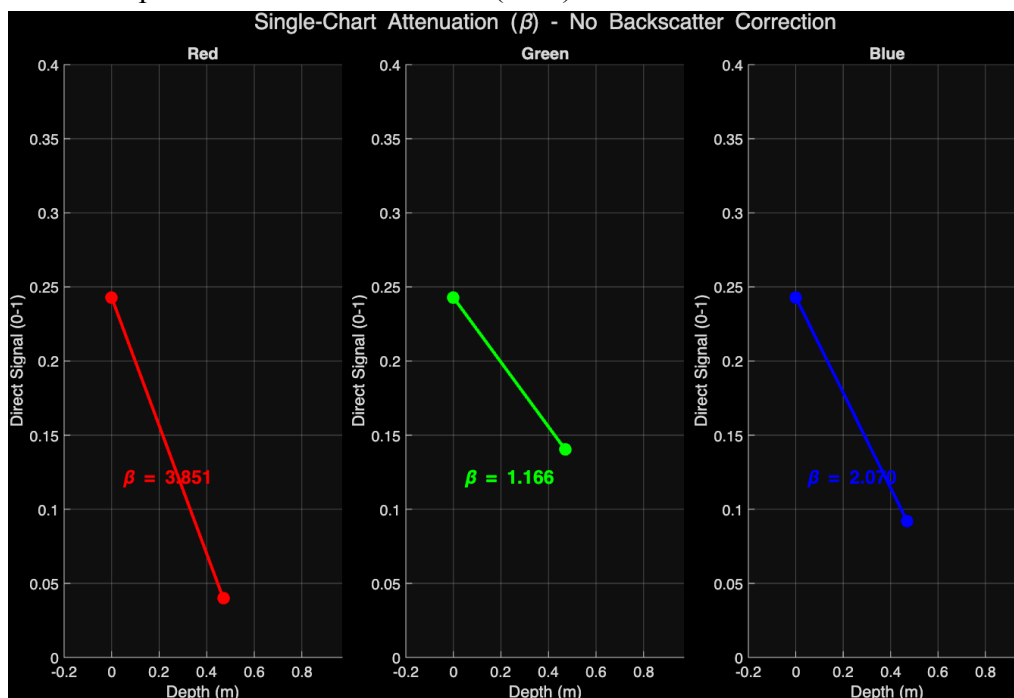
Ex. 3 - Step 2: Global Depth Value Assignment & Assume Backscatter Negligible

- Global depth set at 0.47

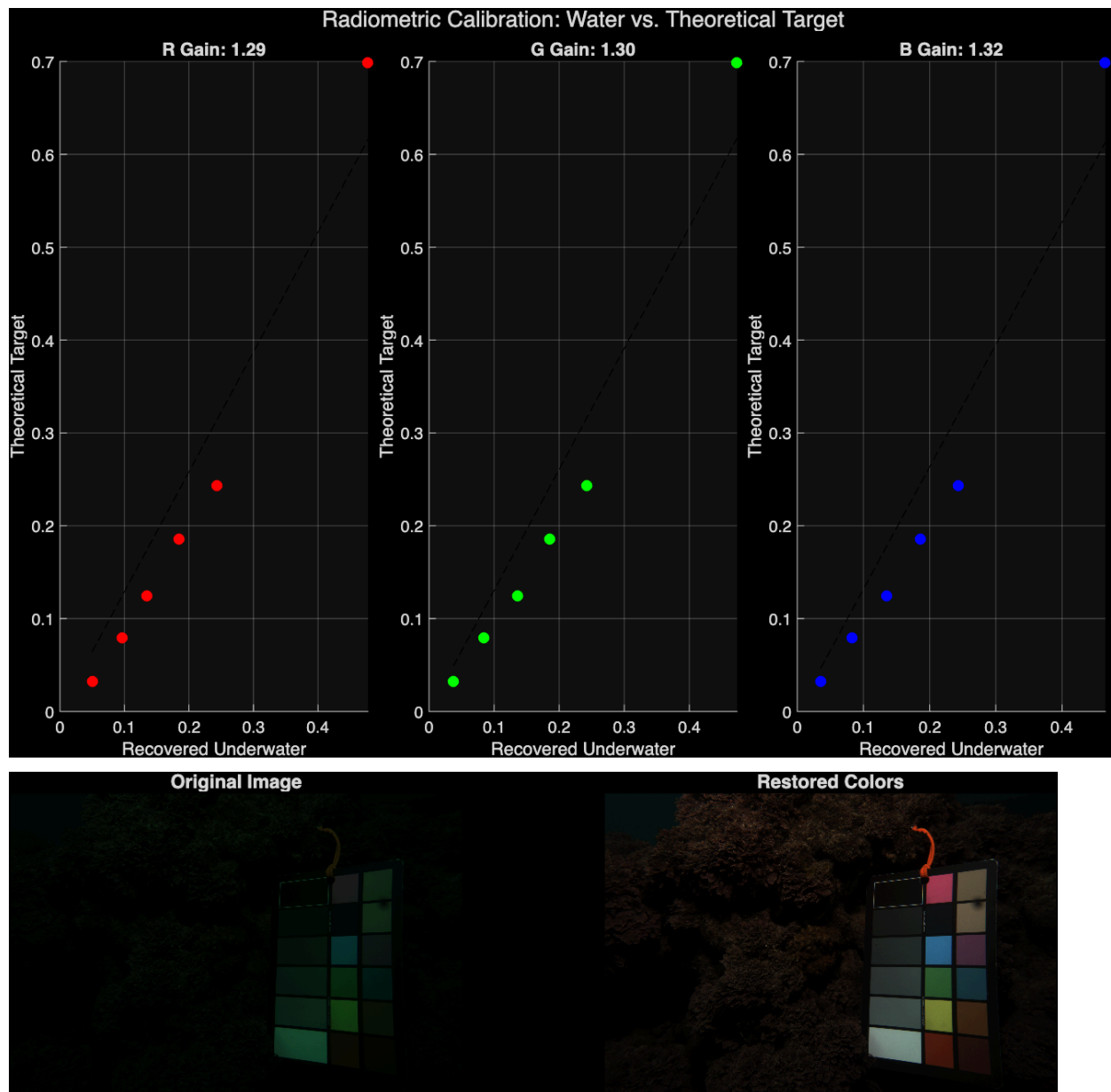
Ex. 3 - Step 3: Backscatter Modeling & Subtraction

- Backscatter assumed to be negligible.

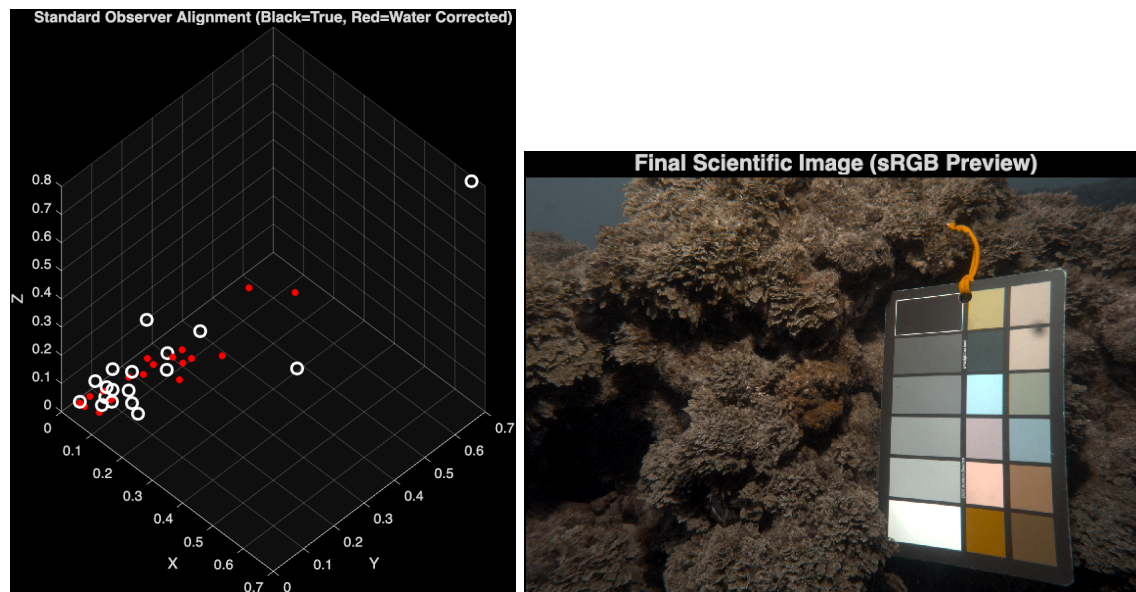
Ex. 3 - Step 4: Wideband Attenuation (Beta) - No-Backscatter Override



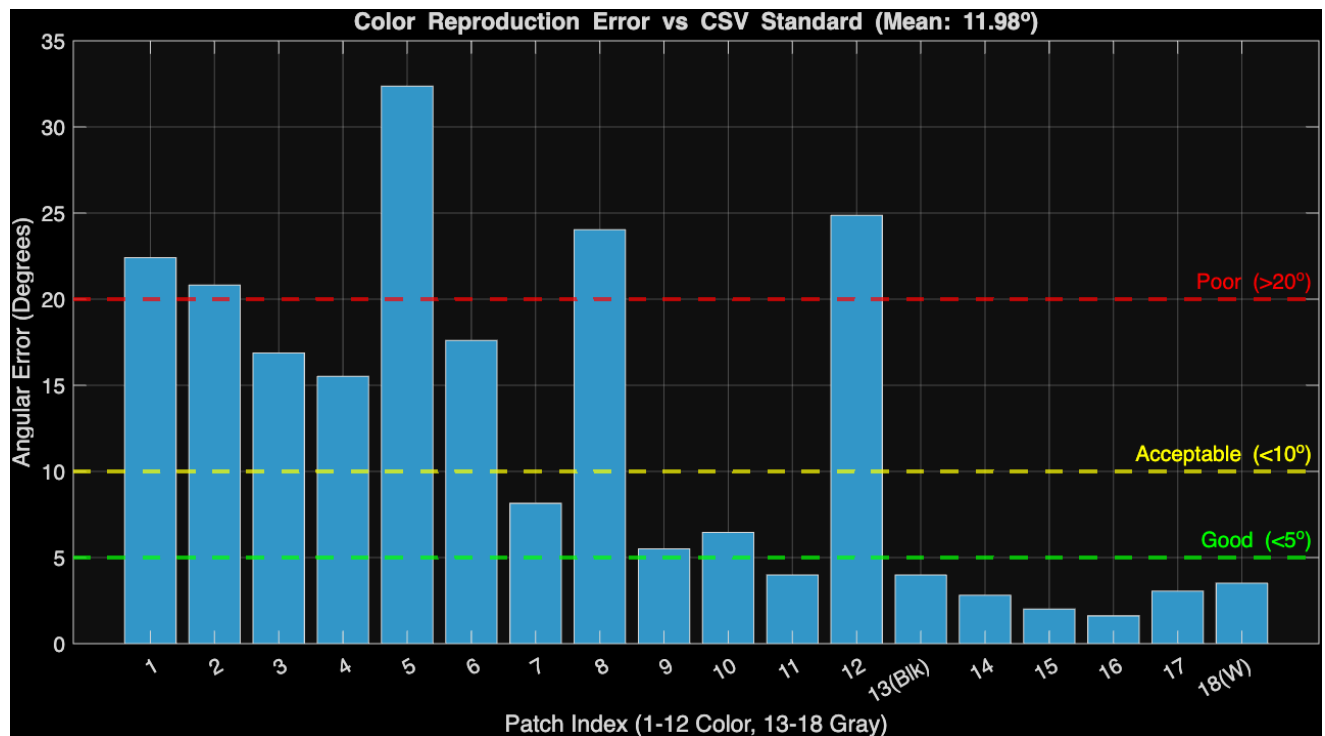
Ex. 3 - Step 5: Attenuation Correction & Radiometric Calibration (Theoretical)



Ex. 3 - Step 6: Universal XYZ Transformation (Water-Corrected -> Standard Observer)



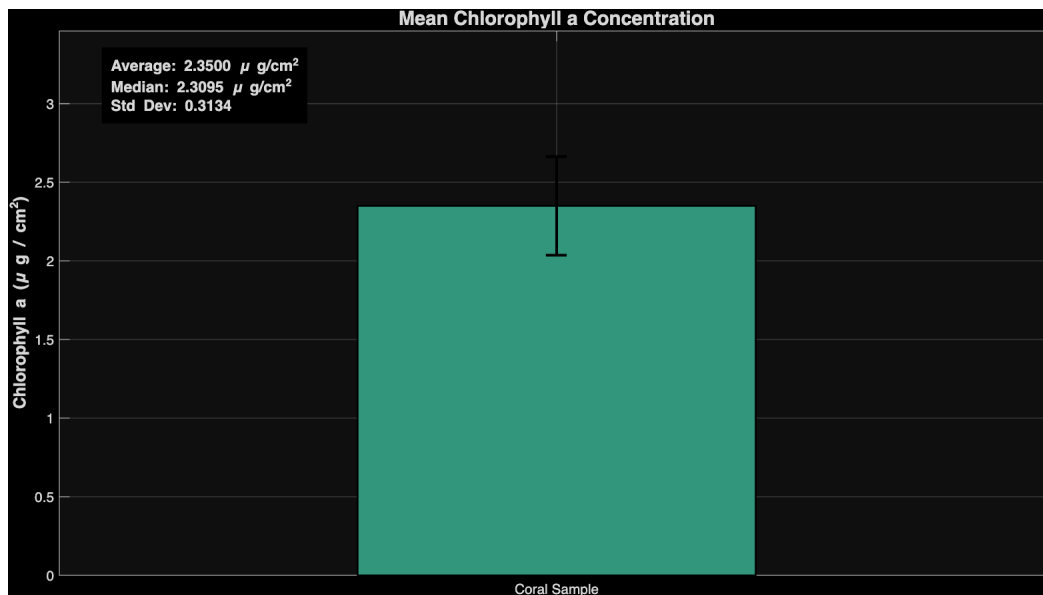
Ex. 3 - Step 7: Color Consistency Error Analysis (Theoretical XYZ)



Ex. 3 - Step 8: Coral Masking and Single-Point Chlorophyll Estimation



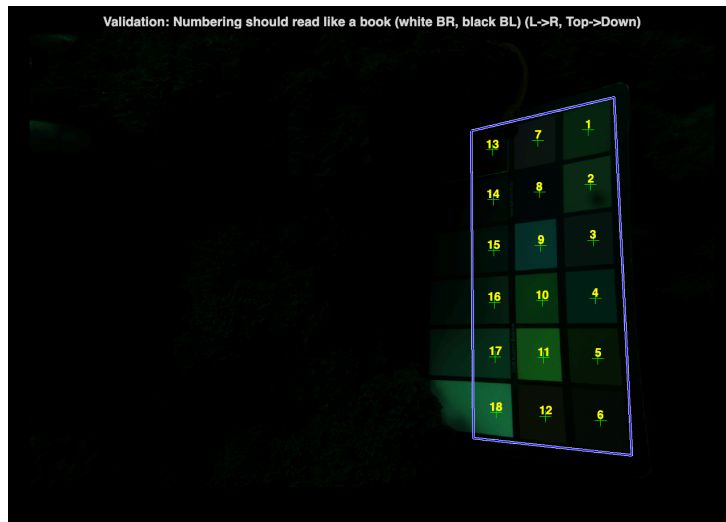
Ex. 3 - Step 9: Chlorophyll Data Extraction & Export



Example #4 - IMG_2280 (Ambient, same aperture, Canon theoretical values)

- Lighting: Ambient
- Script: BA_OculinaPipelineV7_NoBcNoZ_Canon.m
- Depth set globally to 0.4 m (estimated based on other images)
- Backscatter assumed to be negligible.
- Used calculated theoretical XYZ values for the transformation matrix. Theoretical values incorporated the DGK reflectances csv, CIE standard illuminant D65 csv, and **Canon G7 x mk iii spectral signature csv** files for the calculation.
- Final results: **Average angle error = 10.55 degrees**
- Chlorophyll estimation: **Mean = 2.35 $\mu\text{g}/\text{cm}^2$, Std. Deviation = 0.3845**

Ex. 4 - Step 1: Extract Color Chart Data & Create Masks



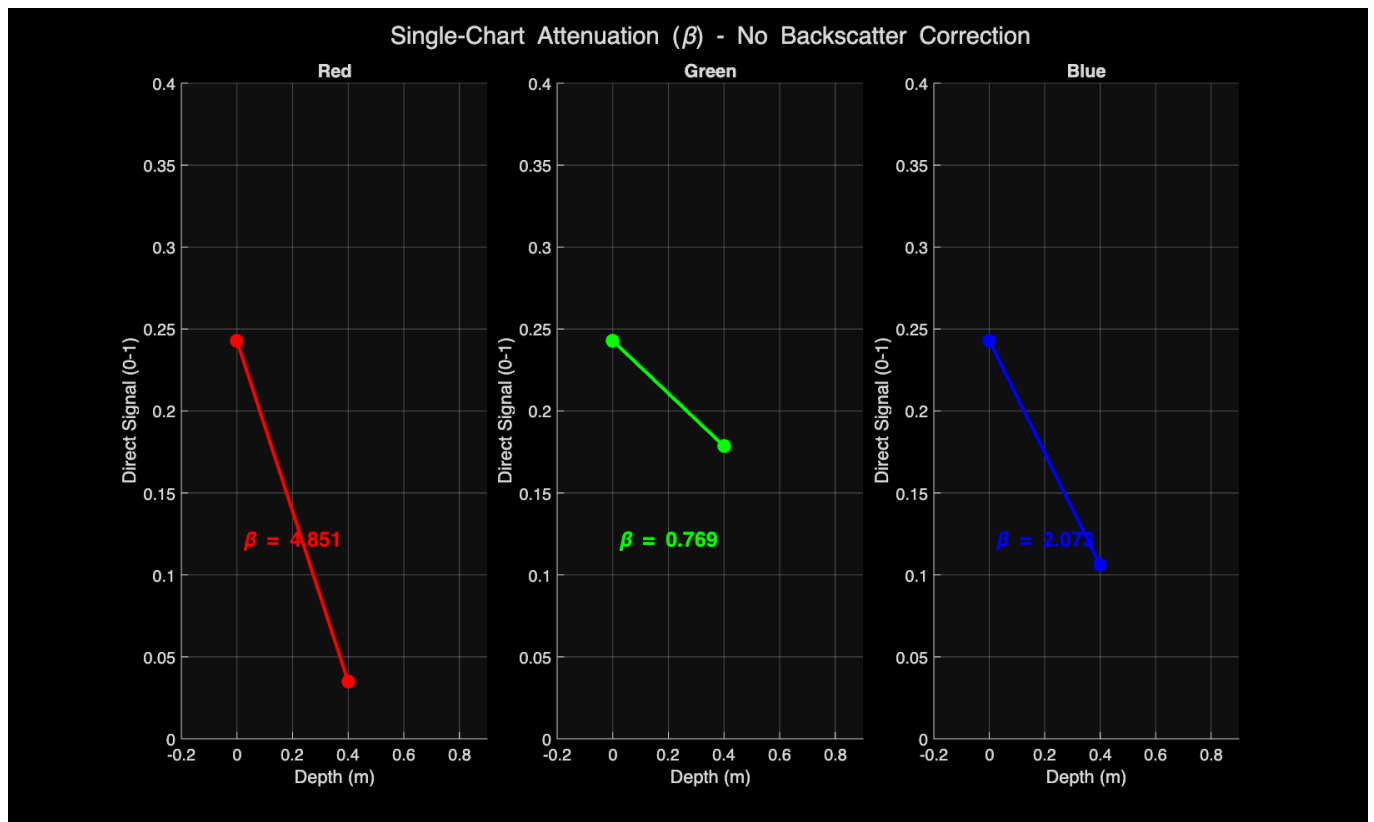
Ex. 4 - Step 2: Global Depth Value Assignment & Assume Backscatter Negligible

- Global depth set at 0.4

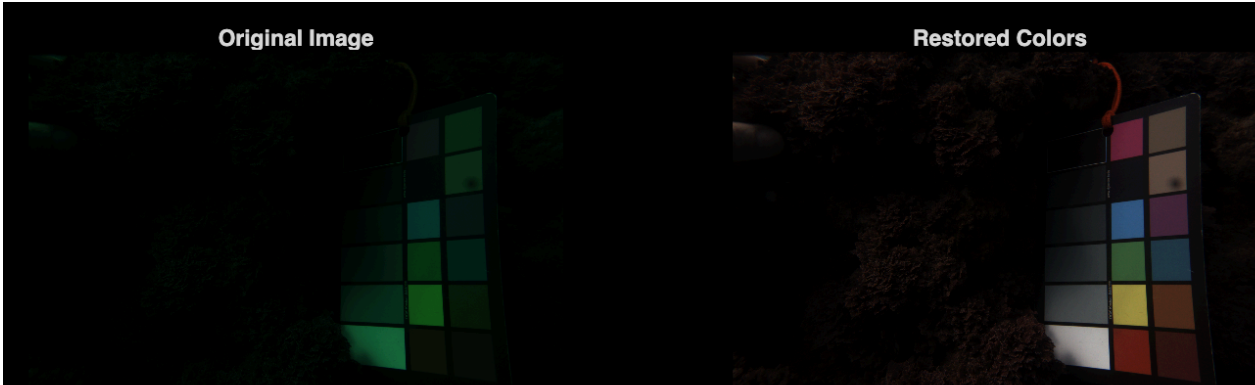
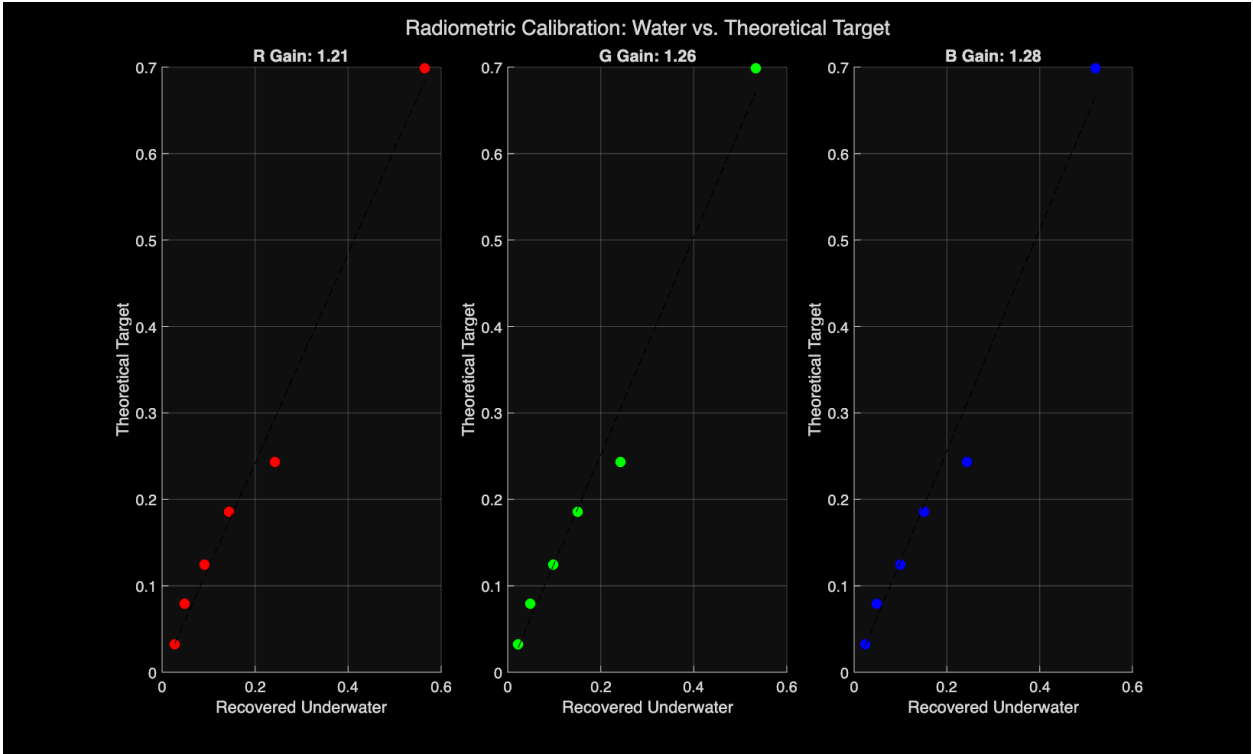
Ex. 4 - Step 3: Backscatter Modeling & Subtraction

- Backscatter assumed to be negligible.

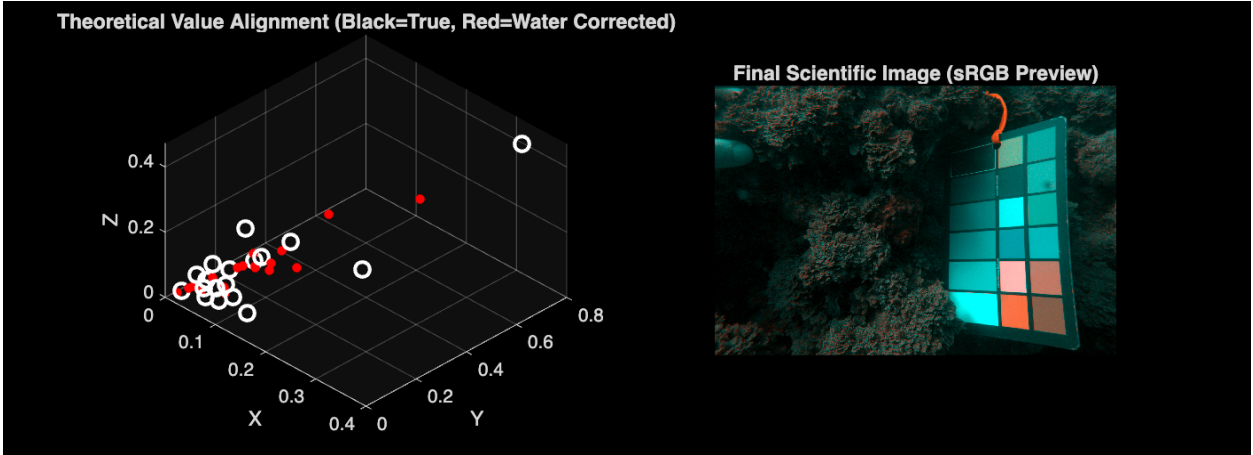
Ex. 4 - Step 4: Wideband Attenuation (Beta) - No-Backscatter Override



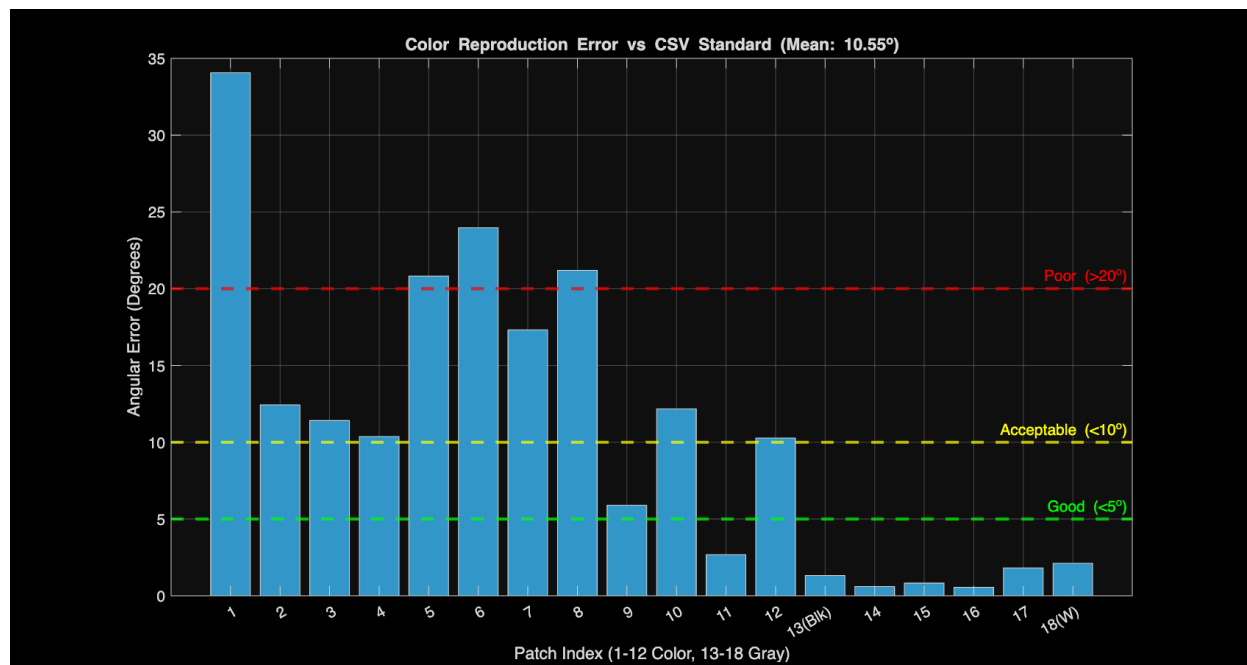
Ex. 4 - Step 5: Attenuation Correction & Radiometric Calibration (Theoretical)



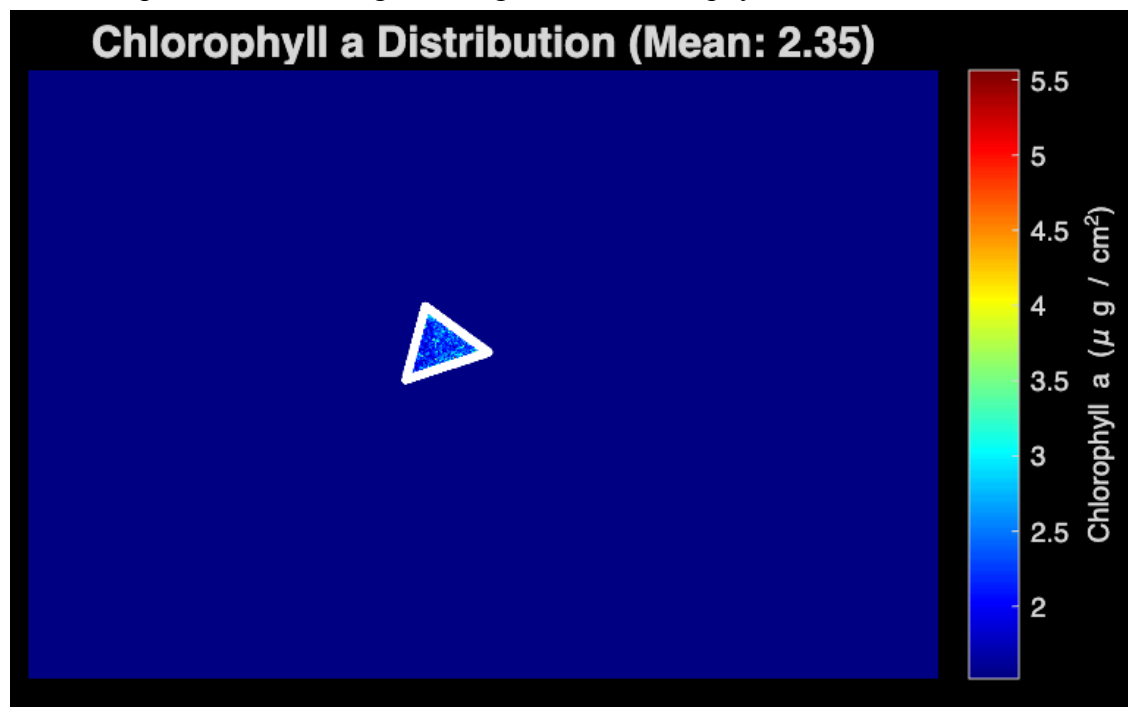
Ex. 4 - Step 6: Universal XYZ Transformation (Water-Corrected -> Theoretical Values)



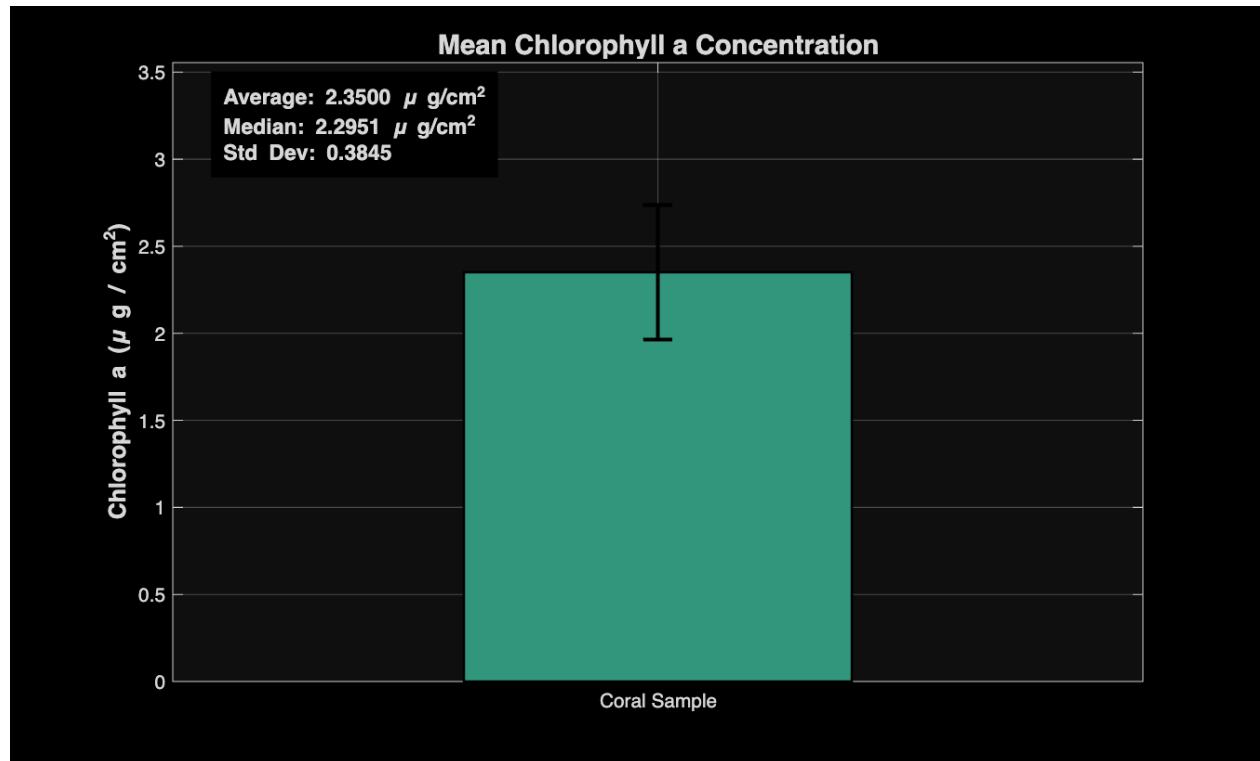
Ex. 4 - Step 7: Color Consistency Error Analysis (Theoretical XYZ)



Ex. 4 - Step 8: Coral Masking and Single-Point Chlorophyll Estimation



Ex. 4 - Step 9: Chlorophyll Data Extraction & Export



Artificial Intelligence Disclosure: This report was written by Boaz Abramson, a human.

Artificial Intelligence was solely utilized for the development of code for the image processing pipeline and for suggestions of minor grammatical adjustments to the final progress report.

Citations

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